Maize (Corn) Diseases

Identification, causes, favourable conditions, management and verified remedies

Eighteen diseases with a Maharashtra focus and all-India context

Primary sources: TNAU Agritech Portal, CIB&RC (Insecticides Act, 1968) Major Uses of Pesticides up to 30.09.2025, ICAR-IIMR / AICRP Maize, MPKV Rahuri and other university extension and research material. Compiled 19 October 2026; cross-checked and extended the same day.

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About this document

Scope and method

Each disease chapter follows the same six headings: identification; causes and progression; favourable conditions; prevention and management; remedies and treatment; and sources. Disease descriptions come from TNAU, ICAR-IIMR/AICRP and MPKV Rahuri material, with university and research papers where noted. Chemical recommendations are included only where they could be matched to the CIB&RC list of registered maize uses or to a published TNAU, MPKV or ICAR recommendation, and each is tagged to show how it was verified.

Verification tags used in the chemical tables

V1 - Verified against CIB&RC. The product is listed for MAIZE and this disease (or a named group including it) in the CIB&RC 'Major Uses of Pesticides - Fungicides' compilation, up to 30.09.2025. Doses are per hectare as printed.

V2 - University recommendation (TNAU or MPKV Rahuri). A matching maize label claim was NOT found in the CIB&RC list in this review; confirm current registration and the product label.

V3 - ICAR / AICRP Maize (IIMR) published state recommendation. Same confirmation step as V2.

-- - Not recommended / no chemical option verified. Sources reviewed do not support a recommendation, so none is given.

Important cautions

- The CIB&RC compilation used for V1 is 'based on certificates issued' up to 30.09.2025 and carries a disclaimer that it is for guidance and not for legal purposes. A hosted copy was read, not the live register. Registrations and labels change; check the current label and the CIB&RC register (ppqs.gov.in) and follow the pre-harvest interval (PHI) printed on the label.
- Doses are per hectare as printed in CIB&RC; TNAU and ICAR advisories are given as concentrations. Water volumes depend on crop stage and sprayer. PHIs are for field maize; check the label for green cobs, baby corn, sweet corn or fodder.
- Two printed doses in the CIB&RC text were unclear or looked inconsistent (azoxystrobin + cyproconazole; metalaxyl-M seed dose) and are flagged where they appear. Where sources give different doses, the label governs.
- Rotate fungicide groups across sprays; use PPE, follow label safety instructions and keep products away from water bodies.
- MPKV Rahuri (AICRP Maize) material was retrieved and used for Maharashtra. No PDKV Akola or UC IPM maize disease page was retrieved, so none is cited. Maharashtra notes are limited to what the cited sources state.
- Sources marked supplementary (Plantix, Nebraska, Pacific fact sheet, review abstracts) support only the specific points where they are cited.

Summary table

Disease	Pathogen / type	Main non-chemical measures	Chemical status
Turcicum Leaf Blight	Exserohilum turcicum (fungal, residue-borne)	Resistant hybrids, stubble destruction, spray on first lesions	Many V1 options (strobilurin/triazole mixes, mancozeb); MPKV 2024 schedule
Maydis Leaf Blight	Bipolaris maydis (fungal, residue-borne)	Resistant hybrids, debris removal, avoid continuous maize	Mancozeb, zineb, several V1 mixes
Banded Leaf and Sheath Blight	Rhizoctonia solani f. sp. sasakii (soil-borne, sclerotia)	Strip lower leaves, drainage, tolerant hybrids	No maize-labelled product found; V3 state advisories only
Common Rust	Puccinia sorghi (fungal; Oxalis alternate host)	Remove Oxalis, destroy residue, tolerant hybrids	Kresoxim-methyl and strobilurin mixes (V1)
Polysora (Southern) Rust	Puccinia polysora (fungal, wind-borne)	Timely sowing, avoid late-maturing hybrids	'Rust' claims on two V1 products; species not named
Downy Mildew (SDM, RDM, BSDM)	Peronosclerospora sorghi and relatives (oomycete)	Seed treatment, rogueing, avoid risky sowing windows	Metalaxyl seed treatments (V1); foliar mancozeb (V1)
Charcoal Rot	Macrophomina phaseolina (soil-borne)	Avoid stress at flowering, potash, rotation, Trichoderma / Pseudomonas	None verified (cultural and biological)
Fusarium Stalk Rot	Fusarium verticillioides complex (soil-, seed-borne)	Potash, clean seed, rotation	None verified; Trichoderma seed treatment (V1, root rot claim)
Bacterial Stalk Rot	Dickeya zeae (bacterial)	Ridges, drainage, avoid waterlogging	Bleaching powder drench (V2/V3)
Curvularia Leaf Spot	Curvularia lunata (seed- and soil-borne)	Seed treatment, tolerant hybrids, avoid diseased residue	Thiram/captan seed treatment (V2); no Curvularia claim in CIB&RC
Brown Spot	Physoderma maydis (chytrid)	Rotation, tolerant varieties, residue clearance	None verified
Zonate Leaf Spot	Gloeocercospora sorghi (fungal)	No sourced advice	None verified
Gray Leaf Spot	Cercospora zeae-maydis / C. zeina (residue-borne)	Resistant hybrids, rotation, residue management	None verified; supplementary sources only
Pythium Stalk Rot	Pythium aphanidermatum (oomycete)	Well-drained fields; avoid dense, high-N stands	None verified
Late Wilt	Harpophora maydis (vascular fungus)	Rotation, debris removal, tolerant hybrids, avoid moisture stress	None verified
Ear Rots / Mycotoxins	Fusarium, Gibberella, Aspergillus, Diplodia, Penicillium	Avoid ear injury and stress; resistant hybrids; cool storage	None verified; supplementary sources only
Common and Head Smut	Ustilago maydis; Sphacelotheca reiliana	Resistant hybrids	None verified; supplementary sources only
Seed Rot, Seedling Blight, Root Rot	Seed- and soil-borne fungal complex	Recommended varieties, avoid cold wet soil at sowing	Several registered seed dressings (V1); Trichoderma (V1)

Quick separation of look-alikes (main diseases): long cigar-shaped grey-green lesions on lower leaves point to turcicum blight; many small vein-limited tan lesions point to maydis blight; concentric bands on lower sheaths with sclerotia point to banded leaf and sheath blight; powdery cinnamon-brown pustules on both surfaces point to common rust, whereas small orange-red pustules mostly on the upper surface point to polysora rust; white-striped, stunted seedlings with downy growth point to downy mildew; pink-red rot inside the stalk points to Fusarium, black shredded stalk with sclerotia to charcoal rot, and foul-smelling wet rot at the base to bacterial stalk rot.

1. Turcicum Leaf Blight (Northern Corn Leaf Blight)

Maize | Exserohilum turcicum (teleomorph Setosphaeria turcica) | Fungal (residue-borne, wind-dispersed)

At a glance. Long grey-green to tan, cigar-shaped lesions that start on lower leaves and can 'burn' the whole canopy. One of the four nationally important maize diseases; listed by MPKV Rahuri as predominant in kharif Maharashtra. Hybrid resistance plus well-timed sprays on first appearance work best.

1. Identification

Symptoms

- Crop / stage: maize at any growth stage; damage is greatest when infection reaches the crop around silking under favourable weather (T3).
- Early lesions are small, slightly oval and water-soaked; they grow into long, elliptical to spindle-shaped, grey-green or tan necrotic lesions, 2.5 to 15 cm long (T3).
- Lesions appear first on the lower leaves and increase in number and size up the plant until the foliage looks scorched; spores form abundantly on both sides of the spot (T3, IIMR).
- TNAU also describes cigar-shaped lesions running parallel to the leaf margin that coalesce, with dusty dark-green/black fuzz (spores) below the lesions and brittle, frost-like leaves (T2).

Distinguishing features

- Maydis leaf blight lesions are smaller and more numerous, mostly rectangular and held between veins; turcicum lesions are larger, fewer and cigar-shaped (IIMR, PPP).
- Common rust gives powdery cinnamon-brown pustules rather than necrotic cigar-shaped lesions (T6).
- Banded leaf and sheath blight shows concentric bands on sheaths and ears with sclerotia, not long lesions on the blade (IIMR).

2. Causes and progression

- Pathogen: Exserohilum turcicum (Pass.) Leonard & Suggs (teleomorph Setosphaeria turcica); brownish septate mycelium; olive-grey, spindle-shaped, curved conidia (1 to 9 septa per TNAU, 4 to 9 per IIMR) (T2, IIMR).
- Survival and primary inoculum: mycelium and conidia in infected crop residue on or in the soil; conidia can thicken into chlamydospores (T2, IIMR). In an Indian study, survival in residue fell from 80% at 3 weeks to 10% at 12 weeks (IIMR).
- Spread: wind-borne conidia from leaf lesions, sometimes over long distances (IIMR). Johnson grass and other grasses are hosts, and Saccharum arundinaceum was identified as an alternate host in Karnataka; the maize isolate does not move to sorghum and vice versa (UAS, IIMR).
- Progression: infection starts on lower leaves, spore production on each lesion feeds the next cycle; severity is highest from tasselling to 6 to 8 weeks after silking, and plants about 65 days old were most vulnerable in Karnataka trials (IIMR, UAS). Physiological races defined by Ht genes exist, so resistance can break down (IIMR).
- Losses: grain yield losses of 25 to 90% are reported in epiphytotics (IIMR); avoidable losses of 52% grain and 40% stover were estimated in a susceptible genotype in Karnataka (UAS).

3. Favourable conditions

- Wet, humid, cool weather later in the crop season (T2).
- IIMR correlates higher disease with temperature 22 to 38 C, relative humidity 72 to 98% and rainfall 134 to 165 mm (IIMR).
- Karnataka epidemiology: minimum 11.3 to 21.4 C, maximum 26.3 to 29.4 C, RH 58.7 to 84.5%; a mid-September sowing had the highest severity while February to April sowings were least affected (UAS). Use only as a pointer; local sowing windows differ.

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Grow resistant or tolerant hybrids; this is the foundation of management (IIMR, PPP). IIMR lists released hybrids rated moderately resistant to TLB, for example DMRH 1301 (recommended for NEPZ) and DMRH 1305 (also MR to maydis leaf blight and resistant to Curvularia leaf spot) (IJA, IHY). Check the AICRP zone and current state release before choosing a hybrid.
- Burn or bury infected maize stubble after harvest (T2); avoid continuous maize in the same field.
- Scout from knee-high stage and start the first spray when lesions first appear (T3, MP24).

- Match the sowing window to local advice: in Karnataka trials, late-September sowings carried far more disease than February to April sowings (UAS).

Maharashtra note

AICRP on Maize, MPKV Rahuri lists turcicum and maydis leaf blight as the predominant kharif diseases of Maharashtra and rust in late kharif; the centre screens maize genotypes against TLB under artificial epiphytotic conditions (MPR). MPKV Rahuri's 2024 recommendation for TLB is two sprays of a combination fungicide (see table), the first on disease appearance and the second 15 days later (MP24).

5. Remedies and treatment

Biological and non-registered options

- Manipur state recommendation: foliar spray of garlic bulb extract at 10% at 35, 45 and 55 days after sowing (ISM). This is a state-specific advisory, not a registered product claim.

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Azoxystrobin 18.2% + Difenoconazole 11.4% SC	Maize blight (TLB) & downy mildew	1 ml/L (0.1%); MPKV: 10 ml per 10 L, two sprays 15 days apart (Manipur: 0.10% at 35 and 45 DAS)	500 L	V1	CIB&RC; PHI 26 d; MPKV Rahuri 2024 (MP24, ISM)
Azoxystrobin 18.2% + Cyproconazole 7.3% SC	Downy mildew, TLB, rust	MPKV: 10 ml per 10 L, two sprays 15 days apart (see note on printed dose)	500 L	V1	CIB&RC; PHI 52 d; MPKV Rahuri 2024 (MP24)
Pyraclostrobin 20% WG	Turcicum leaf blight	100 g a.i. = 500 g/ha	500 L	V1	CIB&RC; PHI 40 d
Kresoxim-methyl 44.3% SC	Turcicum leaf blight, rust	250 g a.i. = 500 ml/ha (TNAU, rust: 1 ml/L)	500 L	V1	CIB&RC; PHI 25 d
Fluxapyroxad 167 g/L + Pyraclostrobin 333 g/L SC	TLB, common rust	150 g a.i. = 300 ml/ha	500 L	V1	CIB&RC; PHI 41 d
Mefentrifluconazole 200 g/L + Pyraclostrobin 200 g/L SC	Northern & southern leaf blight	160-200 g a.i. = 400-500 ml/ha	500 L	V1	CIB&RC; PHI 48 d
Pyraclostrobin 133 g/L + Epoxiconazole 50 g/L SE	Leaf blight	137.25 g a.i. = 750 ml/ha (Manipur: 0.15%)	500 L	V1	CIB&RC; PHI 48 d; ISM
Tebuconazole 50% + Trifloxystrobin 25% WG	Leaf blight	175 + 87.5 g a.i. = 350 g/ha	500 L	V1	CIB&RC; PHI 15 d
Mancozeb 75% WP	Leaf blight (and downy mildew)	1.5-2 kg/ha (TNAU: 2-4 g/L, repeat at 10-day interval; MPKV page: 0.25%)	750 L	V1	CIB&RC; PHI not stated; T3, MPR
Zineb 75% WP	Leaf blight	1.5-2 kg/ha (TNAU: 2-4 g/L)	750-1000 L	V1	CIB&RC; PHI not stated; T3
Carbendazim 12% + Mancozeb 63% WP	Downy mildew & leaf blight	1000 g/ha (120 + 630 g a.i.)	500 L	V1	CIB&RC; PHI 37 d
Mancozeb 75% WP or Dithane Z-78 (zineb)	Leaf blight (E. turcicum named)	2.5 g/L or 3 g/L, 1-2 sprays at 7-10 day interval from knee-high stage	-	V1 / V3	ICAR / AICRP Andhra Pradesh (ISA); mancozeb and zineb have CIB&RC maize leaf-blight claims
Hexaconazole 5% EC	Turcicum leaf blight	1 ml/L at first symptoms; repeat after 15 days	-	V3	ICAR / AICRP Karnataka (ISP); no maize claim for hexaconazole in CIB&RC list
Propiconazole 25% EC	Leaf blight	1 ml/L at 35 and 50 DAS	-	V2	TNAU (T2); maize claim not found in CIB&RC list; HP trial (HPT) supports efficacy

Printed-dose note: the CIB&RC row for azoxystrobin + cyproconazole prints '0.26% a.i. / 1% formulation' for maize. That is ten times the equivalent azoxystrobin + difenoconazole row and does not match MPKV Rahuri's explicit 10 ml per 10 L (1 ml/L). This document uses the MPKV dose and treats the printed '1%' as unreliable; confirm against the product label.

PHIs above are those printed for field maize. Where green cobs, baby corn, sweet corn or green fodder are harvested early, check the label for that use before spraying; PHIs of 25 to 52 days rule out several products.

Rotate fungicide groups (strobilurin, triazole, dithiocarbamate) across the two or three sprays, and follow the label for volume and interval. A Pacific fact sheet based on CABI says chemical control is often uneconomic

and resistance is the main tool (PPP); Indian recommendations (MPKV, TNAU) nonetheless support sprays where the disease is severe.

6. Sources and citations

[T2] TNAU Agritech Portal - Maize: Leaf blight (Exserohilum turcicum & Helminthosporium maydis). https://agritech.tnau.ac.in/crop_protection/maize_disease_new/maize_2.html

[T3] TNAU Agritech Portal - Maize: Turcicum leaf blight (Exserohilum turcicum). https://agritech.tnau.ac.in/crop_protection/maize_disease/maize_4.html

[T6] TNAU Agritech Portal - Maize: Common rust (Puccinia sorghi). https://agritech.tnau.ac.in/crop_protection/maize_disease_new/maize_4.html

[IIMR] Hooda, K.S., Bagaria, P.K., Khokhar, M., Kaur, H. & Rakshit, S. (2018). Mass Screening Techniques for Resistance to Maize Diseases. ICAR-Indian Institute of Maize Research, Ludhiana. https://iimr.icar.gov.in/wp-content/uploads/2020/03/Bulletin-Mass-Screening.pdf

[UAS] Harlapur, S.I. (2007). Epidemiology and management of turcicum leaf blight of maize caused by Exserohilum turcicum. UAS Bangalore (thesis abstract, Krishikosh). https://krishikosh.egranth.ac.in/items/07f2e06e-71b6-4cd4-9672-9b5a754db589

[MPR] AICRP on Maize, Rahuri (Maharashtra) - centre page: predominant diseases and university recommendation. https://aicrpmaize.icar.gov.in/?p=1123

[MP24] Mahatma Phule Krishi Vidyapeeth (MPKV), Rahuri - AICRP on Maize: technological recommendations released (Plant Pathology, 2024). https://mpkv.ac.in/Uploads/Research/22.%20AICRP%20on%20Maize_20250926040628.pdf

[ISM] ICAR-IIMR / AICRP Maize - Manipur state recommendations (turcicum leaf blight). https://iimr.icar.gov.in/?p=1176

[ISA] ICAR-IIMR / AICRP Maize - Andhra Pradesh state recommendations (leaf blight, late wilt, banded leaf and sheath blight, charcoal rot). https://aicrpmaize.icar.gov.in/?p=1163

[ISP] ICAR-IIMR / AICRP Maize - Karnataka state recommendations: disease management for turcicum leaf blight, charcoal stalk rot, common rust and downy mildew (page dated 21-11-2024). https://aicrpmaize.icar.gov.in/?p=1172

[HPT] Management of turcicum leaf blight of maize in Himachal Pradesh (CSK HPKV thesis abstract, KRISHI/Krishikosh). https://krishikosh.egranth.ac.in/handle/1/5810184504

[CIB] Central Insecticides Board & Registration Committee (CIB&RC), Govt. of India: 'Major Uses of Pesticides (Registered under the Insecticides Act, 1968) - Fungicides', up to 30.09.2025, 'based on certificates issued' (Directorate of Plant Protection, Quarantine & Storage). Copy used: https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Earlier edition (31.03.2024) on the official portal: https://ppqs.gov.in/sites/default/files/fungicides_31.03.2024.pdf

[IJA] Newly released high yielding single cross maize hybrids for different ecologies (ICAR-IIMR; Indian Journal of Agricultural Sciences). https://epubs.icar.org.in/index.php/IJAgS/article/download/87102/35591/223006

[IHY] ICAR-IIMR: Catalogue of special features for recently released maize hybrids. https://iimr.icar.gov.in/?p=150

[PPP] Pacific Pests, Pathogens and Weeds (PestNet / CABI-based fact sheet): Maize northern leaf blight (supplementary, non-Indian source). https://apps.lucidcentral.org/ppp/text/web_full/entities/maize_northern_leaf_blight_226.htm

2. Maydis Leaf Blight (Southern Corn Leaf Blight)

Maize | Bipolaris maydis (teleomorph Cochliobolus heterostrophus; syn. Helminthosporium maydis) | Fungal (residue-borne, wind- and rain-dispersed)

At a glance. Many small tan, vein-limited lesions on leaves, caused by a fungus that thrives in warm, wet weather. Moderate to high in four of five Indian maize zones in IIMR surveys; listed as predominant in kharif Maharashtra. Mancozeb sprays are the long-standing recommendation.

1. Identification

Symptoms

- Race O lesions start small and diamond-shaped, elongate to rectangular (2-6 x 3-22 mm), tan, bounded by leaf veins, and stay on the leaves (IIMR).
- Race T lesions are oval and slightly larger (6-12 x 6-27 mm) with dark brown borders; they also occur on stems, sheaths and ears and can cause ear rot, and seedlings from infected seed may wilt and die within 3 to 4 weeks (IIMR).
- Under heavy pressure, especially when infection is before silking, lesions coalesce and blight whole leaves; sugars diverted from the stalk to fill grain can predispose plants to lodging (IIMR).
- TNAU describes both leaf blights together: long, cigar-shaped grey-green to tan lesions on lower leaves, merging to cover the leaf (T2).

Distinguishing features

- Compared with turcicum leaf blight, maydis lesions are smaller, more numerous and mostly rectangular and vein-limited (IIMR, PPP).
- Curvularia leaf spot gives small (about 0.5 cm) lesions with a straw-coloured centre, dark margin and pale halo (IIMR).
- Race T is relevant only where hybrids carry Texas male-sterile cytoplasm (IIMR).

2. Causes and progression

- Pathogen: Bipolaris maydis (Y. Nisik. & C. Miyake) Shoemaker; three races (O, T and C); olive-brown, spindle-shaped, 5 to 11 septate conidia (IIMR).
- Survival: mycelium in infected crop debris left on the soil surface between seasons (IIMR); TNAU adds survival on infected plant debris and wind-blown spores (T2).
- Progression: in warm, humid weather the debris sporulates, conidia are carried by wind and rain splash to young maize, enter through stomata and form lesions that produce more conidia; a disease cycle can be completed in as little as 72 hours (IIMR).
- Losses: close to 70% in warm, wet temperate and tropical areas (IIMR).

3. Favourable conditions

- Warm (20 to 32 C) and damp growing season; long, dry, sunny spells are unfavourable (IIMR).
- Wet, humid, cool weather later in the season, per TNAU's combined leaf-blight page (T2).
- Continuous maize cropping and retained infected debris raise risk (IIMR).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Burn or bury infected maize stubble (T2); rotate and avoid maize after maize where the disease has been severe (IIMR).
- Use resistant or tolerant hybrids. IIMR hybrid notes record moderate resistance to maydis leaf blight in hybrids such as DMRH 1305 and a hybrid rated resistant to MLB (IHY, IJA); confirm suitability for your zone. An older state list names Deccan, VL 42, Prabhat, KH-5901 and Deccan 109 among resistant varieties (ISS), but these are dated.
- Start sprays when lesions first appear and repeat after 10 to 15 days if wet weather continues (ISQ, T2).

Maharashtra note

MPKV Rahuri / AICRP Maize Rahuri lists maydis leaf blight with turcicum leaf blight as predominant in kharif Maharashtra, and its university recommendation is mancozeb 75 WP at 0.25% for leaf blight management (MPR). In IIMR's multi-year survey, MLB severity (PDI) was moderate to high (45 to 65) in four zones and low (25) in the central western zone (PR); local pressure varies by season.

5. Remedies and treatment

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Mancozeb 75% WP	Leaf blight, downy mildew	1.5-2 kg/ha (a.i. 1.125-1.5 kg); IIMR advice: 0.2% (2 g/L), repeat after 10-15 days; Himachal: 2 g/L plus two more sprays at 10-day interval; MPKV page: 0.25%	750 L	V1	CIB&RC; PHI not stated; ISQ, ISS, ISH, MPR
Zineb 75% WP	Leaf blight	1.5-2 kg/ha	750-1000 L	V1	CIB&RC; PHI not stated
Carbendazim 12% + Mancozeb 63% WP	Downy mildew & leaf blight	1000 g/ha	500 L	V1	CIB&RC; PHI 37 d
Azoxystrobin 18.2% + Difenoconazole 11.4% SC	Maize blight & downy mildew	1 ml/L (0.1%)	500 L	V1	CIB&RC; PHI 26 d
Tebuconazole 50% + Trifloxystrobin 25% WG	Leaf blight	350 g/ha (175 + 87.5 g a.i.)	500 L	V1	CIB&RC; PHI 15 d
Pyraclostrobin 133 g/L + Epoxiconazole 50 g/L SE	Leaf blight	750 ml/ha	500 L	V1	CIB&RC; PHI 48 d
Mefentrifluconazole 200 g/L + Pyraclostrobin 200 g/L SC	Southern & northern leaf blight	400-500 ml/ha	500 L	V1	CIB&RC; PHI 48 d
Propiconazole 25% EC	Maydis leaf blight (research)	0.1% (1 ml/L) was among the most effective in a field trial	-	--	Research only (PYT); no maize claim in CIB&RC list; not recommended here

In the RPCAU field trial, propiconazole 25 EC at 0.1% and mancozeb 75 WP at 0.2% gave the lowest disease index (18.5 and 29.6 against 85.2 in untreated plots) (PYT). Only mancozeb has a maize leaf-blight claim in the CIB&RC list.

Follow label PHI and rotate chemical groups across sprays. Sources reviewed give no registered seed or soil treatment specifically for MLB.

6. Sources and citations

[T2] TNAU Agritech Portal - Maize: Leaf blight (Exserohilum turcicum & Helminthosporium maydis). https://agritech.tnau.ac.in/crop_protection/maize_disease_new/maize_2.html

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[PR] ICAR-IIMR: Prevalence of maize diseases in India (KRISHI repository record). https://krishi.icar.gov.in/jspui/handle/123456789/42997

[ISQ] ICAR-IIMR / AICRP Maize - State recommendations: maydis leaf blight, downy mildew, banded leaf and sheath blight, root rot (state not stated on the page). https://aicrpmaize.icar.gov.in/?p=1197

[ISS] ICAR-IIMR / AICRP Maize - State recommendations: turcicum and maydis leaf blight, Fusarium and bacterial stalk rot (state not stated on the page). https://aicrpmaize.icar.gov.in/?p=1179

[ISH] ICAR-IIMR / AICRP Maize - Himachal Pradesh state recommendations (disease management table; page dated 22-11-2024). https://iimr.icar.gov.in/?p=1168

[MPR] AICRP on Maize, Rahuri (Maharashtra) - centre page: predominant diseases and university recommendation. https://aicrpmaize.icar.gov.in/?p=1123

[PYT] Efficacy of newer fungicides against maydis leaf blight disease of maize caused by Helminthosporium maydis (RPCAU; Journal of Pharmacognosy and Phytochemistry 8(6), 2019). https://phytojournal.com/archives/2019.v8.i6.10244/efficacy-of-newer-fungicides-against-maydis-leaf-blight-disease-of-maize-caused-by-ltemgthelminthosporium-maydisltemgt

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https://epubs.icar.org.in/index.php/IJAgS/article/download/87102/35591/223006

[PPP] Pacific Pests, Pathogens and Weeds (PestNet / CABI-based fact sheet): Maize northern leaf blight (supplementary, non-Indian source).
https://apps.lucidcentral.org/ppp/text/web_full/entities/maize_northern_leaf_blight_226.htm

3. Banded Leaf and Sheath Blight

Maize | Rhizoctonia solani f. sp. sasakii (teleomorph Thanatephorus cucumeris) | Fungal (soil- and residue-borne, sclerotia)

At a glance. Concentric, zebra-like bands on lower sheaths and leaves that can climb to the ear and cause ear rot. Soil-borne with sclerotia surviving for years, so it is hard to control with fungicides alone. A nationally important disease, most intense in the hill zone and north-west plains in IIMR surveys.

1. Identification

Symptoms

- Appears on leaves and sheaths of 40 to 50 day old plants and later spreads to the ears (IIMR).
- First seen on the lowest one or two leaf sheaths and leaves as concentric bands and rings: large grey, tan or brown discoloured areas alternating with dark brown bands (IIMR).
- Small round black-brown sclerotia appear in the diseased areas; a light brown cottony mould develops on infected tissue (IIMR).
- The developing ear can be damaged and dry prematurely with cracking of husk leaves; ears may show brown rot with cottony mould and sclerotia (IIMR).

Distinguishing features

- Banded lesions on sheaths with cottony growth and sclerotia separate it from turcicum or maydis leaf blight, which are lesions on the blade (IIMR).
- Bacterial stalk rot has a foul smell and soft slimy stalk rot; BLSB does not (T7, IIMR).
- Brown spot (Physoderma maydis) gives small chlorotic spots arranged in alternating bands across the blade and dark brown circular spots on the midrib (IIMR).

2. Causes and progression

- Pathogen: Rhizoctonia solani Kuhn f. sp. sasakii Exner (anastomosis group AG1-IA); brown mycelium branching at right angles, no asexual spores, loose reddish-brown sclerotia (IIMR). The same group also infects rice, wheat, sorghum, bean and soybean (IIMR).
- Survival: sclerotia and mycelium persist in soil, plant debris and host plants for many years (IIMR).
- Spread: sclerotia move with water, wind and soil; the fungus enters by direct penetration of the cuticle or natural openings and by cell-wall degrading enzymes (IIMR).
- Progression: symptoms start low on the plant at 40 to 50 days, climb up sheaths and leaves, and reach the ear late in the season; losses are often noticed only at harvest (IIMR).
- Losses: 11 to 40% grain yield loss reported, with over 90% in some studies; potential loss up to 100% is cited in a review (IIMR, BLR2).

3. Favourable conditions

- Warm, wet weather; outbreaks typically begin in the early summer months, with frequent rain or irrigation and high temperature (IIMR).
- Humid weather with temperature around 28 C (BLR2).
- Poor drainage and soil compaction, and the presence of susceptible hosts (IIMR).
- Prevalent in the hot, humid Himalayan foothills and the plains; IIMR survey intensity was highest in the northern hill and north-west plain zones (IIMR, PR).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Strip off the lowest 2 to 3 affected leaves together with their sheaths, then spray (Andhra Pradesh recommendation, ISA).
- Improve drainage and avoid waterlogging and compaction (IIMR). Clear crop debris and host weeds, since the fungus overwinters in debris, soil and host plants (IIMR).
- Use tolerant hybrids where available; highly resistant cultivars are scarce, so management relies largely on chemicals plus cultural measures (BLR). IIMR lists some hybrids tolerant to BLSB (IHY); confirm zone suitability.
- Crop rotation helps only partly because the fungus has a wide host range; neither crop rotation nor fungicides alone give control (BLR2, IIMR).

Maharashtra note

No Maharashtra-specific recommendation for BLSB was found in the sources reviewed; MPKV Rahuri's page names turcicum and maydis leaf blight and rust as the predominant diseases (MPR). In IIMR surveys BLSB intensity was highest in the northern hill and north-west plain zones (PR), so treat it as a watch-list disease in Maharashtra and scout humid, high-input fields.

5. Remedies and treatment

Biological and non-registered options

- Neem oil at 2 ml/kg seed as seed treatment and 2 ml/L as spray at disease appearance is an ICAR / AICRP state advisory (ISQ); it is not a CIB&RC maize label claim.
- Research only: Trichoderma harzianum seed treatment combined with a fungicide gave the best reduction (about 54 to 57% control) in SKUAST-Jammu artificial-inoculation trials (SKU). No registered biocontrol claim for BLSB on maize was found in the CIB&RC bio-fungicide list (CIBB).

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Carbendazim 50% WP	BLSB: seed treatment and spray	2 g/kg seed; 2 g/L spray at disease appearance	-	V3	ICAR / AICRP state recommendation (ISQ); no maize claim for carbendazim 50 WP in CIB&RC list
Mancozeb 75% WP	BLSB	2 g/L; give two more sprays at 10-day interval (Himachal Pradesh)	-	V3	ICAR / AICRP Himachal Pradesh (ISH); CIB&RC maize claim is leaf blight and downy mildew, not BLSB
Propiconazole 25% EC	BLSB after stripping lower leaves	1 ml/L	-	V3	ICAR / AICRP Andhra Pradesh (ISA); no maize claim in CIB&RC list
Trifloxystrobin + tebuconazole; fluxapyroxad + difenoconazole	BLSB (laboratory / field research)	-	-	--	Research only (SKU); not recommended here; no BLSB claim on maize in CIB&RC list

No product with an explicit BLSB claim on maize was found in the CIB&RC list. The rows above are ICAR / AICRP state recommendations and carry the V3 caution: confirm that the product is currently registered for maize and follow the label.

Because the pathogen is soil-borne and sclerotia persist, expect partial control only; combine stripping, drainage, hybrid choice and timely sprays.

6. Sources and citations

[T7] TNAU Agritech Portal - Maize: Bacterial stalk rot (Erwinia chrysanthemi pv. zeae). https://agritech.tnau.ac.in/crop_protection/maize_disease/maize_9.html

[IIMR] Hooda, K.S., Bagaria, P.K., Khokhar, M., Kaur, H. & Rakshit, S. (2018). Mass Screening Techniques for Resistance to Maize Diseases. ICAR-Indian Institute of Maize Research, Ludhiana. https://iimr.icar.gov.in/wp-content/uploads/2020/03/Bulletin-Mass-Screening.pdf

[PR] ICAR-IIMR: Prevalence of maize diseases in India (KRISHI repository record). https://krishi.icar.gov.in/jspui/handle/123456789/42997

[BLR] Singh, S.K., Patel, M.B., Thakker, B.N., Hooda, K.S. & Barad, A.K.: Rhizoctonia solani f.sp. sasakii inciting banded leaf and sheath blight of maize and their management: an overview (KRISHI record). https://krishi.icar.gov.in/jspui/handle/123456789/67497

[BLR2] Banded leaf and sheath blight of maize: historical perspectives, current status and future directions (review abstract). https://gender.cgiar.org/publications/banded-leaf-and-sheath-blight-maize-historical-perspectives-current-status-and-future

[SKU] Dar, N.A. (2023). Epidemiology and management of banded leaf and sheath blight caused by Rhizoctonia solani f. sp. sasakii in maize. MSc thesis, SKUAST-Jammu (Krishikosh). https://krishikosh.egranth.ac.in/items/29b1b92d-8bcb-4240-8f1c-331f2f4d312f

[ISQ] ICAR-IIMR / AICRP Maize - State recommendations: maydis leaf blight, downy mildew, banded leaf and sheath blight, root rot (state not stated on the page). https://aicrpmaize.icar.gov.in/?p=1197

[ISH] ICAR-IIMR / AICRP Maize - Himachal Pradesh state recommendations (disease management table; page dated 22-11-2024). https://iimr.icar.gov.in/?p=1168

[ISA] ICAR-IIMR / AICRP Maize - Andhra Pradesh state recommendations (leaf blight, late wilt, banded leaf and sheath blight, charcoal rot). https://aicrpmaize.icar.gov.in/?p=1163

[MPR] AICRP on Maize, Rahuri (Maharashtra) - centre page: predominant diseases and university recommendation. https://aicrpmaize.icar.gov.in/?p=1123

[CIB] Central Insecticides Board & Registration Committee (CIB&RC), Govt. of India: 'Major Uses of Pesticides (Registered under the Insecticides Act, 1968) - Fungicides', up to 30.09.2025, 'based on certificates issued' (Directorate of Plant Protection, Quarantine & Storage). Copy used: https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Earlier edition (31.03.2024) on the official portal: https://ppqs.gov.in/sites/default/files/fungicides_31.03.2024.pdf

[CIBB] CIB&RC: 'Major Uses of Bio-Pesticides (Registered under the Insecticides Act, 1968)', updated up to 30.09.2025 (bio-fungicides). Copy used: https://www.jau.in/storage/document/dynamic_document/residue_bio_pesticide_30092025_20251226_780.pdf

[IHY] ICAR-IIMR: Catalogue of special features for recently released maize hybrids. https://iimr.icar.gov.in/?p=150

4. Common Rust

Maize | Puccinia sorghi | Fungal (obligate rust; wind-borne urediniospores; Oxalis alternate host)

At a glance. Powdery cinnamon-brown pustules on both leaf surfaces in cool, moist weather. AICRP Rahuri lists rust as predominant in late kharif Maharashtra. Spray a strobilurin-based product at disease onset; PHIs on the CIB&RC list run from 25 to 52 days.

1. Identification

Symptoms

- Minute flecks appear on both sides of leaves, followed by circular to elongate, cinnamon-brown, powdery, erumpent pustules on both leaf surfaces (T6, IIMR).
- As the crop matures the pustules turn brownish-black as teliospores form (T6, IIMR).
- In severe cases infection spreads to sheaths, tassels and ears, and leaves yellow and tatter in wind (T6, IIMR).
- Partial resistance shows as chlorotic or necrotic flecks with little sporulation (IIMR).

Distinguishing features

- Polysora (southern) rust pustules are smaller, circular, orange-red to light cinnamon and mostly on the upper leaf surface; common rust pustules are larger, brown to cinnamon-brown and on both surfaces (IIMR).
- Leaf-blight lesions are necrotic spots rather than raised powdery pustules (T3, IIMR).

2. Causes and progression

- Pathogen: Puccinia sorghi Schwein.; heteroecious, with five spore stages; urediniospores are single-celled, echinulate and elliptical, teliospores two-celled (T6, IIMR).
- Alternate hosts: Oxalis species (TNAU names O. europea, O. corniculata, O. stricta); aeciospores from Oxalis are wind-carried to maize, and urediniospores on maize repeat the cycle through the season (T6, IIMR).
- Primary spread through the alternate host; secondary spread by wind-borne urediniospores (T6). Urediniospores do not overwinter in cold northern climates, and teliospores need cold and weathering to break dormancy (IIMR).
- Progression: free water on the leaf is needed; infection needs roughly 3 to 6 hours of leaf wetness, and urediniospores on maize keep the cycle going through the season (IIMR).

3. Favourable conditions

- Cool to warm, moist weather of 15 to 25 C (T6).
- IIMR: 16 to 24 C with heavy dews or relative humidity near 100%; urediniospore germination is best at 15 to 25 C and about 100% RH, and light favours germination (IIMR).
- Staggered or extended sowing keeps young susceptible plants available when spores are in the air (IIMR; described for sweet corn).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Remove alternate hosts (Oxalis) from the field and bunds (T6).
- Collect crop remains after harvest and destroy by burning or burying (T6).
- Grow rust-tolerant hybrids; IIMR's Dharwad and Karnal hot spots are used for rust screening (IIMR). Avoid very staggered sowing in the same area.
- TNAU times foliar sprays at 35 and 50 days after sowing (T6).
- Early sowing (May to June) is advised for common rust in the Karnataka recommendation (ISP).

Maharashtra note

AICRP on Maize, MPKV Rahuri states that rust is a predominant disease in late kharif in Maharashtra (MPR). No MPKV spray schedule specific to rust was found in the sources reviewed.

5. Remedies and treatment

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Kresoxim-methyl 44.3% SC	Turcicum leaf blight, rust	250 g a.i. = 500 ml/ha (TNAU: 1 ml/L at 35 and 50 DAS)	500 L	V1	CIB&RC; PHI 25 d; T6
Azoxystrobin 18.2% + Cyproconazole 7.3% SC	Downy mildew, TLB and rust	MPKV (for TLB): 10 ml per 10 L; CIB&RC printed dose unreliable (see TLB chapter)	500 L	V1	CIB&RC; PHI 52 d; MP24
Fluxapyroxad 167 g/L + Pyraclostrobin 333 g/L SC	TLB, common rust	150 g a.i. = 300 ml/ha	500 L	V1	CIB&RC; PHI 41 d
Tebuconazole (single product)	Rust	1 ml/L at 35 and 50 DAS	-	V2	TNAU (T6); no maize claim for tebuconazole alone in CIB&RC list
Mancozeb 75% WP	Common rust	2 g/L; repeat after 10 days	-	V3	ICAR / AICRP Karnataka (ISP); CIB&RC maize claim for mancozeb is leaf blight and downy mildew
Hexaconazole 5% EC, or azoxystrobin + difenoconazole	Common rust, at 35-50 days after sowing	1 ml/L of either	-	V3	ICAR / AICRP Karnataka (ISP); hexaconazole has no maize claim in CIB&RC list

TNAU also lists chlorothalonil and mancozeb for rust at 35 and 50 DAS, but prints the dose in an ambiguous unit ('2 ml/l' for products that are powders or have no maize rust claim), so no dose is reproduced here. The CIB&RC maize claim for mancozeb 75 WP is leaf blight and downy mildew, not rust.

The CIB&RC list gives the claim as 'Rust' without separating common from polysora rust; check the label wording. Follow PHI, and rotate groups across the two sprays.

6. Sources and citations

[T3] TNAU Agritech Portal - Maize: Turcicum leaf blight (Exserohilum turcicum). https://agritech.tnau.ac.in/crop_protection/maize_disease/maize_4.html

[T6] TNAU Agritech Portal - Maize: Common rust (Puccinia sorghi). https://agritech.tnau.ac.in/crop_protection/maize_disease_new/maize_4.html

[IIMR] Hooda, K.S., Bagaria, P.K., Khokhar, M., Kaur, H. & Rakshit, S. (2018). Mass Screening Techniques for Resistance to Maize Diseases. ICAR-Indian Institute of Maize Research, Ludhiana. https://iimr.icar.gov.in/wp-content/uploads/2020/03/Bulletin-Mass-Screening.pdf

[MPR] AICRP on Maize, Rahuri (Maharashtra) - centre page: predominant diseases and university recommendation. https://aicrpmaize.icar.gov.in/?p=1123

[MP24] Mahatma Phule Krishi Vidyapeeth (MPKV), Rahuri - AICRP on Maize: technological recommendations released (Plant Pathology, 2024). https://mpkv.ac.in/Uploads/Research/22.%20AICRP%20on%20Maize_20250926040628.pdf

[ISP] ICAR-IIMR / AICRP Maize - Karnataka state recommendations: disease management for turcicum leaf blight, charcoal stalk rot, common rust and downy mildew (page dated 21-11-2024). https://aicrpmaize.icar.gov.in/?p=1172

[CIB] Central Insecticides Board & Registration Committee (CIB&RC), Govt. of India: 'Major Uses of Pesticides (Registered under the Insecticides Act, 1968) - Fungicides', up to 30.09.2025, 'based on certificates issued' (Directorate of Plant Protection, Quarantine & Storage). Copy used: https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Earlier edition (31.03.2024) on the official portal: https://ppqs.gov.in/sites/default/files/fungicides_31.03.2024.pdf

5. Polysora (Southern) Rust

Maize | Puccinia polysora | Fungal (obligate rust; wind-borne urediniospores)

At a glance. Small orange-red to cinnamon pustules, thickest on the upper leaf surface, in warm and humid weather. Regionally important in peninsular India and favoured by late-planted, late-maturing maize. Yield losses above 45% have been recorded.

1. Identification

Symptoms

- Pustules are small (0.2 to 2 mm), circular to oval, orange-red to light cinnamon-brown, and usually more numerous on the upper leaf surface; they are slower and less abundant underneath (IIMR).
- Pustules are initially less powdery than common rust because the leaf epidermis stays intact over them for a long time, often opening by a longitudinal slit (IIMR).
- Chocolate-brown to black teliospores later form in a ring around a uredinium (IIMR).

Distinguishing features

- Common rust pustules are larger, brown to cinnamon-brown, on both surfaces and favoured by cooler weather (15 to 25 C); polysora is favoured by warm, humid conditions (IIMR).
- Polysora can infect husks as well as leaves (IIMR).

2. Causes and progression

- Pathogen: Puccinia polysora Underw.; only uredial and telial stages are known; teliospores do not rupture the epidermis, have not been shown to germinate and are unimportant in the cycle (IIMR).
- Spread: wind-blown urediniospores from earlier infected maize, moving progressively through the season; free water as dew is required for germination and penetration (IIMR).
- Progression: symptoms show 3 to 6 days after infection and pustules rupture to expose spores by 7 to 10 days; most rapid at 25 to 28 C with high humidity (IIMR). The fungus mostly survives winter as urediniospores in tropical areas (review summary, PR2).
- Losses: more than 45% in some seasons (IIMR); a review cites 50% or more in outbreaks (PR2).

3. Favourable conditions

- Warm and humid weather; optimum urediniospore germination and infection at 24 to 28 C (IIMR).
- Periods of extended dew or high humidity; heavy rain with high temperature (IIMR).
- Late-planted and late-maturing maize (IIMR).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Avoid late planting and very late-maturing hybrids where polysora rust occurs (IIMR).
- Use hybrids with rust tolerance; IIMR identifies Mandya (Karnataka) as a kharif hot spot for polysora rust screening (IIMR). IIMR's survey found polysora rust at very low infection in the peninsular zone (PR).
- Destroy crop remains after harvest, the same sanitation TNAU advises for common rust (T6); no polysora-specific cultural advice beyond sowing time was found in the sources reviewed.

Maharashtra note

No Maharashtra-specific polysora rust record or spray schedule was found in the sources reviewed. AICRP Rahuri lists 'rust' as predominant in late kharif without naming the species (MPR); the field symptoms in the identification section tell the two rusts apart.

5. Remedies and treatment

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Kresoxim-methyl 44.3% SC	Maize rust (species not named)	250 g a.i. = 500 ml/ha	500 L	V1	CIB&RC; PHI 25 d
Azoxystrobin 18.2% + Cyproconazole 7.3% SC	Maize rust (species not named)	See TLB chapter for dose (MPKV 10 ml per 10 L)	500 L	V1	CIB&RC; PHI 52 d

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Fluxapyroxad + Pyraclostrobin SC	Common rust (named); polysora not named	-	-	--	No polysora-specific claim; not reproduced here

The CIB&RC list names 'Rust' for kresoxim-methyl and azoxystrobin + cyproconazole without distinguishing species; confirm the label wording before using either for polysora rust. No polysora-specific fungicide schedule from ICAR, TNAU or MPKV was found in the sources reviewed, so none is given beyond these registered claims.

Follow PHI and rotate chemical groups.

6. Sources and citations

[IIMR] Hooda, K.S., Bagaria, P.K., Khokhar, M., Kaur, H. & Rakshit, S. (2018). Mass Screening Techniques for Resistance to Maize Diseases. ICAR-Indian Institute of Maize Research, Ludhiana. https://iimr.icar.gov.in/wp-content/uploads/2020/03/Bulletin-Mass-Screening.pdf

[PR] ICAR-IIMR: Prevalence of maize diseases in India (KRISHI repository record). https://krishi.icar.gov.in/jspui/handle/123456789/42997

[PR2] Harnessing genetic resistance in maize and integrated rust management strategies to combat southern corn rust (review, PMC11766486). https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11766486/

[MPR] AICRP on Maize, Rahuri (Maharashtra) - centre page: predominant diseases and university recommendation. https://aicrpmaize.icar.gov.in/?p=1123

[CIB] Central Insecticides Board & Registration Committee (CIB&RC), Govt. of India: 'Major Uses of Pesticides (Registered under the Insecticides Act, 1968) - Fungicides', up to 30.09.2025, 'based on certificates issued' (Directorate of Plant Protection, Quarantine & Storage). Copy used: https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Earlier edition (31.03.2024) on the official portal: https://ppqs.gov.in/sites/default/files/fungicides_31.03.2024.pdf

[T6] TNAU Agritech Portal - Maize: Common rust (Puccinia sorghi). https://agritech.tnau.ac.in/crop_protection/maize_disease_new/maize_4.html

6. Downy Mildew (Sorghum Downy Mildew, with Rajasthan and Brown Stripe types)

Maize | Peronosclerospora sorghi (also P. heteropogoni; Sclerophthora rayssiae var. zeae) | Oomycete (soil-, seed- and air-borne)

At a glance. Systemic seedling disease: pale, striped, stunted plants with white downy growth, often sterile. Plants under a month old are most vulnerable. Seed treatment with metalaxyl-based fungicide, early rogueing and avoiding the risky sowing windows are the main tools.

1. Identification

Symptoms

- Sorghum downy mildew (SDM): infected plants are chlorotic and sometimes stunted with white-striped leaves and abnormal seed set; the chlorotic area includes the base of the blade, with sharp transverse margins between diseased and healthy tissue (T1).
- A white, downy growth appears on both leaf surfaces, more on the lower surface towards the base; tassels can show phyllody; systemically infected plants are usually sterile (T1, IIMR).
- Plants less than a month old are highly susceptible; SDM is reported from Gujarat, Maharashtra, Andhra Pradesh, Karnataka and Tamil Nadu (IIMR).
- Tolerant plants may show systemic symptoms but still set normal seed (T1).

Distinguishing features

- Rajasthan downy mildew (P. heteropogoni): 'half-diseased leaf' with pale bases of the second and third leaf, yellow stripes, plants dying at knee-high stage, and no oospores on maize; oospores form on the collateral grass Heteropogon contortus, which carries the primary inoculum (IIMR).
- Brown stripe downy mildew (Sclerophthora rayssiae var. zeae): narrow, yellowish stripes (3 to 7 mm) bounded by leaf veins, worst on lower leaves, not systemic; losses above 60% recorded in India; seed- and soil-borne (IIMR).

2. Causes and progression

- Pathogen: Peronosclerospora sorghi (W. Weston & Uppal) C.G. Shaw; obligate parasite with coenocytic mycelium; conidia germinate by germ tube; spherical oospores formed in leaf tissue (IIMR).
- Primary inoculum: oospores that survive several seasons in soil; the pathogen is also seed-borne, most readily from freshly harvested or immature seed, though seed dried to below 13% moisture is unlikely to carry viable fungus (IIMR).
- Secondary spread: airborne conidia produced at night on infected leaves infect plants up to about four weeks old, leading to systemic infection (IIMR, T1).
- Losses: more than 70% documented in tropical Asia, up to 90% in Nigeria for SDM (IIMR).

3. Favourable conditions

- Conidia form at 17 to 29 C (optimum 24 to 26 C); germination needs a saturated atmosphere or free water and 21 to 25 C; systemic infection occurs at 11 to 32 C after a wet period of 4 hours or more (IIMR).
- Susceptible seedlings under four weeks old; continuous maize in the same field builds oospores in soil (IIMR).
- ICAR advice: in downy-mildew infected fields, September sowing is not recommended (ISP).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Rogue out affected plants early (T1).
- Grow resistant hybrids; TNAU lists its CO 6 hybrid for downy mildew (T1). Confirm availability and suitability for your zone.
- Avoid monocropping of maize, and avoid September sowing in fields with a downy-mildew history (Karnataka recommendation, ISP).
- Use seed from healthy crops, dried to a safe moisture; seed dried below 13% is unlikely to carry viable fungus (IIMR).
- Research evidence: seed treatment plus foliar sprays of a talc-based Pseudomonas fluorescens, and leaf extracts of Prosopis chilensis or Azadirachta indica, reduced downy mildew in greenhouse and

field trials, but the authors say cost and practicality need more work (KAM). No bio-fungicide claim for maize downy mildew was found in the CIB&RC bio list.

Maharashtra note

SDM is reported from Maharashtra (IIMR). No Maharashtra-specific spray schedule from MPKV Rahuri or PDKV Akola was found in the sources reviewed; the national and TNAU recommendations below apply.

5. Remedies and treatment

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Metalaxyl 35% WS	Sorghum, Philippine, sugarcane and brown stripe downy mildew (seed treatment)	240 g a.i. = 700 g formulation per 100 kg seed (7 g/kg) as slurry; slurry 0.75-1.0 L per 100 kg seed	-	V1	CIB&RC; protection 3.5-4 months depending on variety
Metalaxyl-M 31.8% ES	Downy mildew (seed treatment)	0.76 g a.i. / 2.4 ml formulation as printed; units not printed in the list (numbers fit per kg of seed)	-	V1	CIB&RC; protection 3.5-4 months; confirm dose on the label
Mancozeb 75% WP	Downy mildew and leaf blight (foliar)	1.5-2 kg/ha (Karnataka: 2 g/L at 40-45 DAS; Himachal, brown stripe type: 2 g/L plus two more sprays at 10-day interval)	750 L	V1	CIB&RC; PHI not stated; ISP, ISH
Carbendazim 12% + Mancozeb 63% WP	Downy mildew and leaf blight (foliar)	1000 g/ha	500 L	V1	CIB&RC; PHI 37 d
Azoxystrobin 18.2% + Difenoconazole 11.4% SC	Blight and downy mildew	1 ml/L (0.1%)	500 L	V1	CIB&RC; PHI 26 d
Azoxystrobin 18.2% + Cyproconazole 7.3% SC	Downy mildew, TLB, rust	See TLB chapter (MPKV 10 ml per 10 L)	500 L	V1	CIB&RC; PHI 52 d
Metalaxyl + mancozeb (e.g. Ridomil MZ) / Apron 35 SD	Seed treatment and spray at disease appearance	Seed: 2 g/kg (ISP) or 4 g/kg (ISQ); product strength printed as '4 @ + 64%' in ISP	-	V3	ICAR / AICRP state advisories (ISP, ISQ); doses differ, follow the label
Metalaxyl + mancozeb	Foliar spray at 20 days after sowing	1000 g/ha (as printed in TNAU)	-	V2	TNAU (T1); foliar metalaxyl not a maize claim in CIB&RC list

Seed treatment is the registered route: the CIB&RC maize claims for metalaxyl and metalaxyl-M are seed treatments, and the list gives a protection period of 3.5 to 4 months depending on variety. Sources differ on doses (2, 4 and 7 g per kg of seed for different products and strengths); read the label of the product you buy.

Systemically infected plants are usually sterile (IIMR) and TNAU advises rogueing them (T1).

6. Sources and citations

[T1] TNAU Agritech Portal - Maize: Sorghum downy mildew (Peronosclerospora sorghi). https://agritech.tnau.ac.in/crop_protection/maize_disease/maize_1.html

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[MP24] Mahatma Phule Krishi Vidyapeeth (MPKV), Rahuri - AICRP on Maize: technological recommendations released (Plant Pathology, 2024). https://mpkv.ac.in/Uploads/Research/22.%20AICRP%20on%20Maize_20250926040628.pdf

7. Charcoal Rot (Post-Flowering Stalk Rot)

Maize | Macrophomina phaseolina | Fungal (soil- and residue-borne, sclerotia)

At a glance. Stalks of hot, dry-stressed maize turn grey-black and shred inside after flowering, then lodge. Worst under drought and high soil temperature, so rabi and moisture-stressed crops are most at risk. Management is mainly cultural (water at flowering, potash, rotation) plus Trichoderma or Pseudomonas.

1. Identification

Symptoms

- Appears 1 to 2 weeks after flowering: plants wilt, the outside of the lower internode turns straw-coloured and the pith disintegrates (T5, T4).
- Inside the stalk there is black discolouration and shredding of vascular bundles, with small greyish-black sclerotia visible on the rind and bundles; stalks often break at the crown (T5, T4).
- Premature ripening and drying of the plant; infected kernels may blacken (T5, T4).

Distinguishing features

- Fusarium stalk rot shows reddish or pink discolouration inside the stalk rather than black shredded tissue with sclerotia (T8).
- Bacterial stalk rot gives a wet, foul-smelling basal rot that kills plants around tasselling (T7).
- Pythium stalk rot is confined to the first internode above ground, water-soaked and twisted, before flowering (IIMR).

2. Causes and progression

- Pathogen: Macrophomina phaseolina; round to irregular black sclerotia; pycnidia on stalks with colourless single-celled spores (T4).
- Primary spread: sclerotia in infected crop and soil debris; secondary spread by wind-borne pycnidiospores (T4). The fungus invades seedling roots and moves up the stem as the plant matures (T5).
- Progression: symptomless root infection early, then stalk colonisation under stress after flowering, leading to weak stalks and lodging (T5, KHO).
- Losses: post-flowering stalk rots as a complex cause 10 to 42% loss and up to 100% in some areas (KHO).

3. Favourable conditions

- Dry and hot weather during and after flowering (T4).
- Soil temperature 30 to 42 C and low soil moisture (T4, T5); low soil pH (5.4 to 6.0) (T4).
- Water or nutrient stress at flowering and grain filling (T4, ISP).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Avoid water stress at flowering; irrigate at flowering and grain filling and avoid monocropping of maize (T4, ISP). The Karnataka recommendation adds balanced fertiliser with potash and rotation with paddy or sesame (ISP).
- Crop rotation, removal of plant debris, summer ploughing and tolerant hybrids (ISA).
- Avoid nutrient stress; apply potash at 80 kg/ha in endemic areas (T4).
- Soil application of Pseudomonas fluorescens or Trichoderma viride at 2.5 kg/ha mixed with 50 kg well-decomposed FYM (mixed 10 days before use), at 30 days after sowing (T4); an older TNAU page says to apply Trichoderma in furrows with FYM (T5).
- Seed treatment per ICAR: 25 g PSB (Pseudomonas striata H-21) plus 6 g Trichoderma harzianum per kg seed (ISP).
- IIMR lists released hybrids rated moderately resistant to charcoal rot, for example DMRH 1301 and DMRHP 1402 (IJA); check zone suitability.

Maharashtra note

MPKV Rahuri's AICRP centre is developing a sick plot to screen maize genotypes against charcoal rot in rabi, and tests hybrids against it in rabi trials (MPR), which signals that rabi maize in Maharashtra is exposed. No MPKV spray or soil recommendation for charcoal rot was found.

5. Remedies and treatment

Biological and non-registered options

- Soil application of P. fluorescens or T. viride at 2.5 kg/ha with 50 kg FYM at 30 DAS (TNAU, T4) - university recommendation.
- Seed treatment with Trichoderma harzianum 2.0% WP (NBRI-1055) at 20 g/kg seed is on the CIB&RC bio-fungicide list for maize root rot (Fusarium moniliforme) and Fusarium wilt; charcoal rot is not named (CIBB).

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Thiram 40% FS (Thiram Flo)	Charcoal stalk rot (ICAR seed treatment)	5 ml/kg seed (ICAR); CIB&RC maize claim is seedling blight at 24 ml per 10 kg seed	-	V3	ICAR / AICRP (ISP); dose differs from label claim, follow the label (CIB)
Foliar fungicides	Charcoal rot	-	-	--	No registered foliar claim found; stalk damage has occurred by the time symptoms show; not recommended

No fungicide with a charcoal-rot claim on maize was found in the CIB&RC list. Rely on water management, nutrition, rotation and the biological options above.

6. Sources and citations

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[T7] TNAU Agritech Portal - Maize: Bacterial stalk rot (Erwinia chrysanthemi pv. zeae). https://agritech.tnau.ac.in/crop_protection/maize_disease/maize_9.html

[T8] TNAU Agritech Portal - Maize: Fusarium stalk rot (Fusarium moniliforme). https://agritech.tnau.ac.in/crop_protection/maize_disease/maize_10.html

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8. Fusarium Stalk Rot (Post-Flowering Stalk Rot)

Maize | Fusarium verticillioides (syn. F. moniliforme; Gibberella fujikuroi complex) | Fungal (soil-, residue- and seed-borne)

At a glance. Wilting and dull green leaves after flowering, with pink to reddish rot and shredded pith inside the lower stalk. Part of the post-flowering stalk-rot complex with charcoal rot. Potash, rotation, clean seed and avoiding stress are the supported measures; there is no curative fungicide.

1. Identification

Symptoms

- Plants wilt, leaves change from light to dull green, and the lower stalk becomes straw-coloured (T8).
- Reddish discolouration inside the stalk; pith disintegrates leaving mainly vascular bundles (T8).
- If infection occurs just after flowering, husks look bleached and straw-coloured (T8).
- Indian surveys also describe drooping and drying leaves, empty cobs and a wider angle between stalk and cob (FRO).

Distinguishing features

- Charcoal rot has black shredded tissue with sclerotia (T4); Fusarium rot is pink to red.
- Gibberella stalk rot can also look red inside and is easily confused with Fusarium stalk rot (UNL).

2. Causes and progression

- Pathogen: Fusarium verticillioides and related species of the Fusarium fujikuroi complex; Indian isolates from 40 sites in five zones were mostly F. verticillioides (FRO).
- Spread: the fungus enters through roots and grows into the lower stem (T8); it survives as mycelium in debris on or in the soil and can be seed-borne (UNL, T8).
- Wounds from insects or hail help entry (UNL). Fusarium often occurs together with other stalk rots (UNL).
- Losses: 10 to 42% for the post-flowering stalk-rot complex, up to 100% in some areas (KHO).

3. Favourable conditions

- Low potassium: potassic fertiliser reduces infection (T8). The disease is part of the post-flowering stalk-rot complex that follows stress around flowering (KHO).
- Dry spells early in the season followed by heavy rain mid-season (UNL; Nebraska data, indicative only).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Apply potassic fertiliser; it reduces infection (T8).
- Do not plant seed from infected areas; rotate with other crops (T8).
- Older resistant varieties named by TNAU are Ranjit and Ganga 5 (T8); use current tolerant hybrids from IIMR/AICRP lists, for instance MR to Fusarium stalk rot in IIMR's catalogue (IHY).

Maharashtra note

No Maharashtra-specific Fusarium stalk rot recommendation was found in the sources reviewed.

5. Remedies and treatment

Biological and non-registered options

- Trichoderma harzianum 2.0% WP (NBRI-1055): seed treatment at 20 g/kg seed; CIB&RC lists it for maize root rot (Fusarium moniliforme) and Fusarium wilt (CIBB).

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Carbendazim 25% + Mancozeb 50% WS	Maize seed rot, seedling blight (seed dressing)	7.5 + 15 g a.i. = 30 g per 10 kg seed	-	V1	CIB&RC; seed treatment; not a stalk-rot claim
Foliar or soil fungicides	Fusarium stalk rot	-	-	--	None verified; not recommended

The seed dressing above addresses seed rot and seedling blight, not established stalk rot. TNAU lists no chemical for Fusarium stalk rot (T8).

6. Sources and citations

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[T8] TNAU Agritech Portal - Maize: Fusarium stalk rot (Fusarium moniliforme). https://agritech.tnau.ac.in/crop_protection/maize_disease/maize_10.html

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[UNL] University of Nebraska-Lincoln CropWatch: Fusarium stalk rot (supplementary, non-Indian source). https://cropwatch.unl.edu/plant-disease/corn/fusarium-stalk-rot/

[IHY] ICAR-IIMR: Catalogue of special features for recently released maize hybrids. https://iimr.icar.gov.in/?p=150

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9. Bacterial Stalk Rot

Maize | Dickeya zeae (syn. Erwinia chrysanthemi pv. zeae) | Bacterial (residue- and water-associated)

At a glance. Fast, foul-smelling, water-soaked rot at the base of the stalk before or around tasselling, after heavy rain and heat. Reported mainly from the northern hill and north-west plain zones. Drainage, ridges and a pre-flowering bleaching-powder drench are the university measures.

1. Identification

Symptoms

- Symptoms appear on plants 40 to 60 days old: tips of upper leaves wither and dry, then lower leaves yellow (IIMR).
- The stalk near the ground becomes water-soaked and brown, soft and easily breakable; plants die shortly after tasselling (T7, IIMR).
- Rotting tissue gives a foul smell and may carry fly larvae; the stalk interior becomes a soft mass and plants topple (T7, IIMR).

Distinguishing features

- Foul smell and soft wet rot separate it from fungal stalk rots (charcoal rot is dry and shredded; Fusarium is pink and dry) (T4, T8, T7).

2. Causes and progression

- Pathogen: Dickeya zeae (older names Erwinia chrysanthemi pv. zeae); it also infects rice, sugarcane, sorghum, potato and tomato (IIMR).
- Overwintering in stalk residues above the soil, for up to about a year (PLX).
- Progression: rapid decay of stalk tissue once infection starts; IIMR records incidence up to 80 to 85% in nature and 98.8% yield loss in artificial epiphytotics (IIMR).

3. Favourable conditions

- High temperature and humidity (IIMR); intermittent heavy rain followed by hot, humid weather (PLX).
- Waterlogging and poor drainage (T7); flooding or heat may produce symptoms first around the plant base (PLX).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Plant on ridges, avoid waterlogging and drain fields well (T7).
- Apply recommended nitrogen and potassium and use resistant varieties where available (ISH).
- Remove infected stalk residue after harvest, since the bacterium overwinters there (PLX).

Maharashtra note

No Maharashtra record for bacterial stalk rot was found; in IIMR surveys it was restricted to the northern hill and north-west plain zones (PR).

5. Remedies and treatment

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Bleaching powder (33% chlorine)	Bacterial stalk rot, soil drench	10 kg/ha at pre-flowering stage	-	V2	TNAU (T7); also ICAR state advisory (ISS)
Streptocycline / antibiotics	Bacterial stalk rot	-	-	--	No maize claim in CIB&RC list; not recommended

Bleaching powder is a university recommendation rather than a CIB&RC pesticide label; apply as advised and keep it off foliage. Some trade articles suggest captan with bleaching powder, but that was not found in a university or CIB&RC source and is not included.

6. Sources and citations

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[T7] TNAU Agritech Portal - Maize: Bacterial stalk rot (Erwinia chrysanthemi pv. zeae). https://agritech.tnau.ac.in/crop_protection/maize_disease/maize_9.html

[T8] TNAU Agritech Portal - Maize: Fusarium stalk rot (Fusarium moniliforme). https://agritech.tnau.ac.in/crop_protection/maize_disease/maize_10.html

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[ISS] ICAR-IIMR / AICRP Maize - State recommendations: turcicum and maydis leaf blight, Fusarium and bacterial stalk rot (state not stated on the page). https://aicrpmaize.icar.gov.in/?p=1179

[PLX] Plantix crop-disease library: bacterial stalk rot of maize (supplementary, non-university source). https://plantix.net/en/library/plant-diseases/300035/bacterial-stalk-rot-of-maize/

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10. Curvularia Leaf Spot

Maize | Curvularia lunata (also C. pallescens, C. eragrostidis) | Fungal (seed- and soil-borne)

At a glance. Small round to oval spots with a straw-coloured centre, dark brown margin and pale halo in hot, humid weather. IIMR calls it a very important seed- and soil-borne disease of hot, humid maize areas, with damage up to 60% reported. No fungicide with a Curvularia claim on maize is in the CIB&RC list.

1. Identification

Symptoms

- Small necrotic or chlorotic spots with a light-coloured halo; fully developed lesions are about 0.5 cm across with a straw-coloured to light brown centre and a dark brown margin (IIMR).
- TNAU: flecking that develops into larger round to oval lesions, separate or coalescing, 1 to 6 mm across, with a straw to light brown centre and dark brown margin (T9).
- Spots can cover the whole leaf and blight it; in a Karnataka survey severity was highest from tasselling to post-flowering (CLSK, CLSR).

Distinguishing features

- Maydis leaf blight lesions are tan, rectangular and limited by veins, without the pale halo and dark margin (IIMR).
- Zonate leaf spot gives circular or target-shaped lesions (IIMR).

2. Causes and progression

- Pathogen: Curvularia lunata (Wakker) Boedijn; a Karnataka survey found C. lunata most common, followed by C. eragrostidis; TNAU also names C. pallescens (IIMR, CLSK, T9).
- Seed- and soil-borne (IIMR); a Rajasthan study demonstrated seed transmission and assessed a 22.29% yield loss (CLSR).
- Wide host range including rice, pearl millet, soybean, groundnut, cotton and pulses (IIMR).
- Overwinters in corn residue and is splashed or wind-blown to new plants (UTC, supplementary).
- Losses: damage up to 60% from loss of photosynthetic area is cited by IIMR; Karnataka severity ranged from 20.5 to 63.6% (IIMR, CLSK).

3. Favourable conditions

- Hot, humid climate from flowering to grain filling (CLSK).
- Hot, humid maize areas generally (IIMR); continuous maize and conservation tillage raise risk in a US guide (UTC, supplementary).
- IIMR hot spots: Dhaulakuan (Himachal Pradesh) and Udaipur (Rajasthan) (IIMR).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Seed treatment with thiram or captan at 2 g/kg seed reduces seed infection (T9).
- Avoid planting into residue from a diseased crop (UTC, supplementary); rotate away from maize.
- Use tolerant hybrids: DMRH 1305 is reported resistant to Curvularia leaf spot (IJA).
- A Jilin (China) study found potassium fertiliser lowered infection probability while nitrogen and phosphorus did not (JIL, supplementary).

Maharashtra note

No Maharashtra-specific record or recommendation was found in the sources reviewed.

5. Remedies and treatment

Biological and non-registered options

- TNAU also lists captafol for this disease; captafol does not appear in the CIB&RC list, so it is not reproduced.

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Thiram (seed treatment)	Curvularia leaf spot, seed infection	2 g/kg seed	-	V2	TNAU (T9); CIB&RC maize claim for thiram is seedling blight only

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Captan (seed treatment)	Curvularia leaf spot, seed infection	2 g/kg seed	-	V2	TNAU (T9); no maize claim for captan in CIB&RC list
Mancozeb 75 WP, or hexaconazole 5 EC, or azoxystrobin + difenoconazole	TLB spray schedule said to help manage Curvularia too	2.5 g/L, or 1 ml/L, or 1 ml/L at first symptoms; repeat after 15 days	-	V3	ICAR / AICRP Karnataka (ISP); hexaconazole has no maize claim in CIB&RC list
Tebuconazole 50% + Trifloxystrobin 25% WG	Curvularia leaf spot (field trial)	0.1% foliar spray was effective in a Karnataka trial	-	--	Research only (CLSK); CIB&RC maize claim is leaf blight (350 g/ha, PHI 15 d), not Curvularia

No product with a Curvularia claim on maize was found in the CIB&RC list. The ICAR Karnataka schedule is a state advisory and carries the V3 caution.

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11. Brown Spot

Maize | Physoderma maydis | Fungus-like (chytrid; obligate parasite; resting sporangia)

At a glance. Small yellow spots in bands across the leaf and dark brown spots on the midrib, in high-rainfall, high-temperature areas. Recorded in Maharashtra among other Indian states. Yield loss is usually modest. No registered fungicide; management is cultural.

1. Identification

Symptoms

- First symptoms on leaf blades: small chlorotic spots arranged as alternate bands of diseased and healthy tissue (IIMR).
- Spots on midribs are circular and dark brown; nodes and internodes show brown lesions; severe infection can induce stalk rot and lodging (IIMR).
- Attacks leaves, leaf sheaths, stalks and sometimes outer husk (IIMR).

Distinguishing features

- Banded leaf and sheath blight has concentric bands on sheaths and sclerotia, not small chlorotic spots with dark midrib spots (IIMR).
- Rusts give raised powdery pustules (IIMR).

2. Causes and progression

- Pathogen: Physoderma maydis (Miyabe) Miyabe; an obligate parasite forming thick-walled resting sporangia that survive in corn debris or soil (IIMR).
- Spread: sporangia move by air currents, insects, splashing rain and flowing water; free water in the whorl or behind sheaths lets sporangia release 20 to 50 zoospores that infect young tissue (IIMR).
- Progression: symptoms and new sporangia appear about 6 to 20 days after infection (IIMR).
- In India recorded from Maharashtra, West Bengal, Rajasthan, Himachal Pradesh and Kashmir; reported losses range from about 5 to 10%, up to 27% in artificial inoculations of susceptible hybrids (IIMR).

3. Favourable conditions

- High rainfall and high mean temperature; sporangia need free water on leaf surfaces and temperatures of 23 to 30 C (IIMR).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Crop rotation; grow resistant or tolerant varieties; clean the field of crop residue (Himachal Pradesh recommendation, ISH).

Maharashtra note

IIMR lists Maharashtra among the Indian states where brown spot has been found (IIMR). No Maharashtra-specific management advice was found.

5. Remedies and treatment

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Foliar or soil fungicides	Brown spot	-	-	--	No maize claim in the CIB&RC list and no verified recommendation; not recommended

No fungicide recommendation for brown spot appears in the ICAR state advisories reviewed or in the CIB&RC list; rely on rotation, residue clearance and tolerant varieties.

6. Sources and citations

[IIMR] Hooda, K.S., Bagaria, P.K., Khokhar, M., Kaur, H. & Rakshit, S. (2018). Mass Screening Techniques for Resistance to Maize Diseases. ICAR-Indian Institute of Maize Research, Ludhiana. https://iimr.icar.gov.in/wp-content/uploads/2020/03/Bulletin-Mass-Screening.pdf

[ISH] ICAR-IIMR / AICRP Maize - Himachal Pradesh state recommendations (disease management table; page dated 22-11-2024). https://iimr.icar.gov.in/?p=1168

[CIB] Central Insecticides Board & Registration Committee (CIB&RC), Govt. of India: 'Major Uses of Pesticides (Registered under the Insecticides Act, 1968) - Fungicides', up to 30.09.2025, 'based on certificates issued' (Directorate of Plant Protection, Quarantine & Storage). Copy used: https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Earlier edition (31.03.2024) on the official portal: https://ppqs.gov.in/sites/default/files/fungicides_31.03.2024.pdf

12. Zonate Leaf Spot

Maize | Gloeocercospora sorghi (syn. Microdochium sorghi) | Fungal

At a glance. Circular, target-shaped leaf lesions in humid, cloudy weather, reported on maize and sorghum in India with up to 85% of photosynthetic leaf area damaged. Mostly a forage and leaf-area concern. No registered fungicide or sourced management in the material reviewed.

1. Identification

Symptoms

- Small non-diagnostic lesions first appear on the lower leaves, anywhere on the blade (IIMR).
- Mature lesions become circular or target-shaped in the leaf interior and semicircular at the margins; lesions elongate, run together and can blight whole leaves (IIMR).
- In favourable weather the disease moves up the plant to all leaves (IIMR).

Distinguishing features

- Curvularia leaf spot lesions are small (about 0.5 cm) with a halo (IIMR).
- Maydis and turcicum blights are tan or grey-green elongated lesions, not targets (IIMR).

2. Causes and progression

- Pathogen: Gloeocercospora sorghi D.C. Bain & Edgerton, forming pink to salmon spore masses and black sclerotia in necrotic tissue (IIMR).
- Hosts include sorghum, pearl millet, sugarcane, Johnson grass, Sudan grass, Bermuda grass and maize (IIMR).
- Impact: in India it damaged up to 85% of photosynthetic leaf area under humid, cloudy weather and reduced leaf weight, which affects forage production (IIMR).

3. Favourable conditions

- Humid and cloudy weather conditions (IIMR).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- The sources reviewed give no management advice for this disease. Sanitation and rotation are general measures, but they are not tested for zonate leaf spot in the material retrieved.

Maharashtra note

No Maharashtra-specific record was found.

5. Remedies and treatment

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Foliar or soil fungicides	Zonate leaf spot	-	-	--	No maize claim in the CIB&RC list and no verified recommendation; not recommended

IIMR's bulletin describes screening only; no registered fungicide or ICAR / TNAU spray advice was found.

6. Sources and citations

[IIMR] Hooda, K.S., Bagaria, P.K., Khokhar, M., Kaur, H. & Rakshit, S. (2018). Mass Screening Techniques for Resistance to Maize Diseases. ICAR-Indian Institute of Maize Research, Ludhiana. https://iimr.icar.gov.in/wp-content/uploads/2020/03/Bulletin-Mass-Screening.pdf

[CIB] Central Insecticides Board & Registration Committee (CIB&RC), Govt. of India: 'Major Uses of Pesticides (Registered under the Insecticides Act, 1968) - Fungicides', up to 30.09.2025, 'based on certificates issued' (Directorate of Plant Protection, Quarantine & Storage). Copy used: https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Earlier edition (31.03.2024) on the official portal: https://ppqs.gov.in/sites/default/files/fungicides_31.03.2024.pdf

13. Gray Leaf Spot

Maize | Cercospora zeae-maydis and C. zeina | Fungal (residue-borne)

At a glance. Long, narrow, rectangular grey to tan lesions between veins, worst in warm, humid, high-dew conditions. A yield-limiting disease of highland Asia, but it is not among IIMR's national or regional priority diseases, so this chapter rests on international and supplementary sources. No maize claim in CIB&RC.

1. Identification

Symptoms

- Small brown or tan spots, often with a yellow halo, on lower leaves usually before flowering; they turn grey, enlarge into elongated rectangular lesions parallel to the veins and can engulf whole leaves (GL1).
- Mature lesions are grey to tan, sharply rectangular, long and narrow (GL4).
- Heavy infection before grain filling can cause large yield loss and weakened stalks that lodge (GL1).

Distinguishing features

- Turcicum blight lesions are cigar-shaped, not narrow and rectangular (IIMR).
- Yellow streaks running along the veins are streak virus, not this fungus (GL2).

2. Causes and progression

- Pathogen: Cercospora zeae-maydis (two sibling groups) and Cercospora zeina (GL3, GL4).
- Survives in maize residue for long periods; wind carries spores from residue to lower leaves, then upper leaves (GL1, GL2).
- Early infection leads to the highest losses (GL2).

3. Favourable conditions

- Temperatures of 25 to 30 C, high humidity with dews or fog, and prolonged leaf wetness; hot, dry weather slows it (GL1).
- High relative humidity, minimum tillage and maize monoculture in highland Asia (GL3).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Grow resistant or tolerant hybrids; a review for Asia calls tolerance breeding the economical, sustainable route (GL3).
- Rotate maize with legumes or other non-host crops and reduce residue (GL2).
- Because the fungus survives on residue left on the soil surface, clearing or burying residue is a logical measure (GL2). In South African trials, though, fungicides managed the disease better than tillage aimed at cutting inoculum (GL5).
- Hybrids differ in susceptibility, but no commercial hybrid in the South African response trial was resistant; the long-term answer is resistant hybrids (GL5).
- Timing evidence: sprays started when the disease had reached the basal five leaves, before the exponential phase of the epidemic, gave the best control and yield (GL5).

Maharashtra note

No Maharashtra-specific information and no ICAR / TNAU / MPKV recommendation for gray leaf spot was found in the sources reviewed.

5. Remedies and treatment

Biological and non-registered options

- No biological control is available for this disease according to Plantix (GL1, supplementary).

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Strobilurin- or triazole-based foliar products	Gray leaf spot	International advisories name pyraclostrobin, azoxystrobin + propiconazole and similar mixes (GL1)	-	--	No gray leaf spot claim for maize in the CIB&RC list; not recommended here

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Azoxystrobin + cyproconazole	Gray leaf spot and white spot (Brazil trials)	Reduced disease and raised yield in twelve single-cross hybrids	-	--	Research only (GL6); the CIB&RC maize label lists downy mildew, TLB and rust, not gray leaf spot
Propiconazole or mancozeb	Gray leaf spot (US trials)	Significantly reduced severity in each experiment	-	--	Research only (GL7); mancozeb has a CIB&RC maize claim for leaf blight and downy mildew, not gray leaf spot; propiconazole has no maize claim

Several registered maize products (see Appendix A) cover other leaf diseases. None lists gray leaf spot, so their use against it would be off-label. The trial evidence above shows what works elsewhere, not what is registered in India. No Indian (ICAR, TNAU, state university) gray leaf spot recommendation or survey was found despite repeated searches.

6. Sources and citations

[IIMR] Hooda, K.S., Bagaria, P.K., Khokhar, M., Kaur, H. & Rakshit, S. (2018). Mass Screening Techniques for Resistance to Maize Diseases. ICAR-Indian Institute of Maize Research, Ludhiana. https://iimr.icar.gov.in/wp-content/uploads/2020/03/Bulletin-Mass-Screening.pdf

[GL1] Plantix crop-disease library: Grey leaf spot of maize (supplementary, non-university source). https://plantix.net/plant-disease/en/100107/grey-leaf-spot-of-maize/

[GL2] Plantwise factsheet for farmers: Grey leaf spot disease management in maize (Zambia, 2013; supplementary). https://factsheetadmin.plantwise.org/Uploads/PDFs/20147801404.pdf

[GL3] Dhami, N.B., Shrestha, J. & Rijal, T.R. (2015). A review on threat of gray leaf spot disease of maize in Asia. Journal of Maize Research and Development 1(1): 71-85. https://www.nepjol.info/index.php/JMRD/article/view/14245

[GL4] Suresh, L.M. et al. (2025). Identification of gray leaf spot-resistant donor lines in tropical maize germplasm. Frontiers in Plant Science (supplementary, CIMMYT Africa). https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2025.1536981/epub

[GL5] Epidemiology and management of grey leaf spot: a new disease of maize in South Africa (UKZN thesis; supplementary, non-Indian). https://researchspace.ukzn.ac.za/signposting/describedby/e77f399f-f4cd-4d18-a88c-15069ecf9eee

[GL6] Brito, A.H., Von Pinho, R.G., Pereira, J.L.A.R. & Balestre, M. (2013). Chemical control of gray leaf spot, white spot and grain rot in maize. Revista Ceres 60(5): 629-635 (supplementary, Brazil). https://www.redalyc.org/pdf/3052/305228952005.pdf

[GL7] Probabilities for profitable fungicide use against gray leaf spot in hybrid maize (Plant Disease; WorldWideScience record, supplementary, US). https://worldwidescience.org/topicpages/g/gray+leaf+spot.html

[CIB] Central Insecticides Board & Registration Committee (CIB&RC), Govt. of India: 'Major Uses of Pesticides (Registered under the Insecticides Act, 1968) - Fungicides', up to 30.09.2025, 'based on certificates issued' (Directorate of Plant Protection, Quarantine & Storage). Copy used: https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Earlier edition (31.03.2024) on the official portal: https://ppqs.gov.in/sites/default/files/fungicides_31.03.2024.pdf

14. Pythium Stalk Rot

Maize | Pythium aphanidermatum | Oomycete (soil- and water-borne)

At a glance. A pre-flowering rot of the first internode above ground in hot, humid, badly drained fields. Damage is usually patchy. No registered chemical; the sources identify the wet, dense, high-nitrogen conditions that favour it.

1. Identification

Symptoms

- Rot is usually confined to the first internode above the soil: stalks become soft, water-soaked, darkened and collapsed (IIMR).
- The stalk twists at the site of infection and lodges, but seldom breaks off completely; plants can look healthy for a while because vascular bundles remain intact (IIMR).
- It can occur before flowering (IIMR).

Distinguishing features

- Bacterial stalk rot gives a foul-smelling, soft, water-soaked basal rot at 40 to 60 days (IIMR, T7).
- Charcoal and Fusarium stalk rots appear after flowering (T4, T8).

2. Causes and progression

- Pathogen: Pythium aphanidermatum (Edson) Fitzp.; survives in soil as mycelium or oospores for several years (IIMR).
- Oospores form sporangia that release swimming zoospores, which infect at or just below the soil surface where free water is present; flooding and excess wetness move the pathogen (IIMR).
- Damage is usually localized because natural dispersal is limited; whole parts of a field may occasionally lodge (IIMR).

3. Favourable conditions

- Hot, humid regions with poorly drained soils; valley and river-bottom fields and depressions are most at risk (IIMR).
- High temperatures (25 to 35 C), high crop density and high nitrogen fertiliser favour severity (IIMR).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Choose well-drained fields and avoid depressions and river-bottom plots, which IIMR identifies as high risk (IIMR).
- IIMR lists high plant density and high nitrogen as favouring severity, so avoiding both is a logical but untested measure in the sources reviewed.

Maharashtra note

No Maharashtra-specific record was found.

5. Remedies and treatment

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Foliar or soil fungicides	Pythium stalk rot	-	-	--	No maize claim in the CIB&RC list and no verified recommendation; not recommended

No registered product or ICAR recommendation for Pythium stalk rot was found; IIMR's bulletin addresses screening for resistance.

6. Sources and citations

[IIMR] Hooda, K.S., Bagaria, P.K., Khokhar, M., Kaur, H. & Rakshit, S. (2018). Mass Screening Techniques for Resistance to Maize Diseases. ICAR-Indian Institute of Maize Research, Ludhiana. https://iimr.icar.gov.in/wp-content/uploads/2020/03/Bulletin-Mass-Screening.pdf

[T4] TNAU Agritech Portal - Maize: Charcoal rot (Macrophomina phaseolina). https://agritech.tnau.ac.in/crop_protection/maize_disease_new/maize_3.html

[T7] TNAU Agritech Portal - Maize: Bacterial stalk rot (Erwinia chrysanthemi pv. zeae). https://agritech.tnau.ac.in/crop_protection/maize_disease/maize_9.html

[T8] TNAU Agritech Portal - Maize: Fusarium stalk rot (Fusarium moniliforme). https://agritech.tnau.ac.in/crop_protection/maize_disease/maize_10.html

[CIB] Central Insecticides Board & Registration Committee (CIB&RC), Govt. of India: 'Major Uses of Pesticides (Registered under the Insecticides Act, 1968) - Fungicides', up to 30.09.2025, 'based on certificates issued' (Directorate of Plant Protection, Quarantine & Storage). Copy used: https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Earlier edition (31.03.2024) on the official portal: https://ppqs.gov.in/sites/default/files/fungicides_31.03.2024.pdf

15. Late Wilt

Maize | Harpophora maydis (syn. Cephalosporium maydis, Magnaporthiopsis maydis) | Fungal (vascular; soil- and seed-borne)

At a glance. Sudden wilting from the top of the plant around flowering, with dark vascular bundles, caused by a vascular fungus that survives long in soil. IIMR lists it among six regionally important diseases; described as emerging in India, with Karnataka studies and an Andhra Pradesh advisory. Resistant hybrids and cultural measures are the options; no registered fungicide.

1. Identification

Symptoms

- No symptoms until about tasselling; wilting then starts from the top leaves and the plant dries rapidly (LW4); first symptoms appear at roughly 50 to 60 days (LW3).
- Drying moves up the plant and stem rotting follows; stem discolouration spreads between internodes (LW2).
- Secondary invaders often make stem symptoms more severe (LW3).

Distinguishing features

- Charcoal rot has black shredded pith with sclerotia; late wilt shows vascular discolouration and a sudden dry-down (LW2, T4).
- Bacterial stalk rot is wet and foul-smelling (T7).

2. Causes and progression

- Pathogen: Harpophora maydis, a vascular fungus first identified in Egypt and reported in India in 1970 (LW3, LW5).
- Survival and spread: persists for long periods in soil; several sources also describe it as seed-borne (LW2, LW5), while the CGIAR genebank note says it is seed-borne but not seed-transmitted (LW4), so sources differ.
- Part of the post-flowering stalk-rot complex (LW1).
- Losses: up to 51% reported in India and incidence of about 70% in Indian fields (LW1, LW2); a Karnataka study confirmed yield loss from artificial inoculation (LW1).

3. Favourable conditions

- The sources reviewed do not give specific weather thresholds; ICAR advice links late wilt with moisture stress after flowering (ISA).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Crop rotation, removing plant debris, summer ploughing, avoiding moisture stress after flowering and growing tolerant hybrids (Andhra Pradesh recommendation, ISA).
- Resistant or tolerant hybrids are the main control worldwide; virulent variants may threaten them (LW5).

Maharashtra note

No Maharashtra-specific record was found. Karnataka studies (Chikkaballapur) and the Andhra Pradesh state advisory show where it is reported (LW1, ISA).

5. Remedies and treatment

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Azoxystrobin seed coating	Late wilt (Israel research)	Not a registered maize use in India	-	--	Research only (LW3); not recommended here

No fungicide with a late-wilt claim on maize is in the CIB&RC list.

6. Sources and citations

[IIMR] Hooda, K.S., Bagaria, P.K., Khokhar, M., Kaur, H. & Rakshit, S. (2018). Mass Screening Techniques for Resistance to Maize Diseases. ICAR-Indian Institute of Maize Research, Ludhiana. https://iimr.icar.gov.in/wp-content/uploads/2020/03/Bulletin-Mass-Screening.pdf

[LW1] Assessment of impact of late wilt caused by Harpophora maydis on maize (UAS Bangalore, Mysore Journal of Agricultural Sciences, 2020). https://old.uasbangalore.edu.in/images/2020-2nd-Issue/3.pdf

[LW2] Modified method of screening maize inbred lines to late wilt disease caused by Harpophora maydis (UAS Bangalore, 2016). https://mail.uasbangalore.edu.in/images/2016-4th-Issue/2.pdf

[LW3] Degani, O. et al. (2019). Evaluating azoxystrobin seed coating against maize late wilt disease using a sensitive qPCR-based method. Plant Disease (MIGAL copy). https://www.migal.org.il/system/files/2022-07/Degani%20et%20al%202019%20%28H.m.%20Seed%20dressing%29%20-%20Plant%20Disease%20-%20with%20Links.pdf

[LW4] CGIAR Crop Genebank Knowledge Base: Maize fungi - late wilt, black bundle disease of maize (Acremonium / Cephalosporium maydis). https://cropgenebank.sgrp.cgiar.org/index.php/153-general-management/stog-maize?start=4

[LW5] Drori, R. et al. (2013). Molecular diagnosis for Harpophora maydis, the cause of maize late wilt in Israel. Phytopathologia Mediterranea 52(1): 16-29 (MIGAL record). https://www.migal.org.il/en/node/5449

[ISA] ICAR-IIMR / AICRP Maize - Andhra Pradesh state recommendations (leaf blight, late wilt, banded leaf and sheath blight, charcoal rot). https://aicrpmaize.icar.gov.in/?p=1163

[T4] TNAU Agritech Portal - Maize: Charcoal rot (Macrophomina phaseolina). https://agritech.tnau.ac.in/crop_protection/maize_disease_new/maize_3.html

[T7] TNAU Agritech Portal - Maize: Bacterial stalk rot (Erwinia chrysanthemi pv. zeae). https://agritech.tnau.ac.in/crop_protection/maize_disease/maize_9.html

[CIB] Central Insecticides Board & Registration Committee (CIB&RC), Govt. of India: 'Major Uses of Pesticides (Registered under the Insecticides Act, 1968) - Fungicides', up to 30.09.2025, 'based on certificates issued' (Directorate of Plant Protection, Quarantine & Storage). Copy used: https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Earlier edition (31.03.2024) on the official portal: https://ppqs.gov.in/sites/default/files/fungicides_31.03.2024.pdf

16. Ear Rots and Mycotoxin Risk (Fusarium, Gibberella, Aspergillus, Diplodia, Penicillium)

Maize | Fusarium verticillioides, Gibberella zeae, Aspergillus flavus, Stenocarpella maydis, Penicillium spp. | Fungal (residue-borne; spread by insects, wind, rain)

At a glance. Fungal rots of the ear that reduce grain quality and can load grain with fumonisin, aflatoxin or vomitoxin. IIMR screens for four ear rots, but the detail here comes from Indian research (PAU) and extension guides from other countries. No registered chemical; management is cultural and about avoiding ear injury and stress.

1. Identification

Symptoms

- Fusarium ear rot: kernels streaked or discoloured, infection can occur anywhere on the ear, often with a 'starburst' pattern (ER4).
- Gibberella ear rot: a red mat of mycelium that may turn white, usually from the ear tip toward the base (ER4).
- Diplodia (Stenocarpella) ear rot: a thick white mat of mycelium, usually from the base toward the tip (ER4).
- Aspergillus ear rot: olive-green to yellow-green powdery spores at the tip or at wounds; produces aflatoxin (ER2, ER4).
- Penicillium ear rot ('blue eye'): blue-green growth, most often on damaged ears (ER2).

Distinguishing features

- Banded leaf and sheath blight ear damage has a cottony light-brown mould with black sclerotia (IIMR).
- Maydis leaf blight race T can also cause ear rot (IIMR).

2. Causes and progression

- Fusarium verticillioides (fumonisin) and F. graminearum (Gibberella zeae; vomitoxin) are the main Fusarium causes; Aspergillus flavus makes aflatoxin; Stenocarpella maydis causes Diplodia rot (ER2, ER3).
- Fungi persist in soil and crop debris and spread by insects, wind and rain to the ears (ER3).
- Fusarium conidia infect maize silks, particularly in tropical regions; F. verticillioides, F. proliferatum and F. equiseti are associated with ear rot in North India, and a fumonisin toxicosis outbreak in India is cited (ER1).
- Corn may carry mycotoxins even when no ear rot is visible (ER3).

3. Favourable conditions

- Ear injury is the first risk factor listed for ear rots and grain moulds (ER2).
- Other stressful conditions are listed with ear injury as risk factors in the Nebraska guide (ER2).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Reduce the infectious fungus in the environment through good irrigation timing and nutrient management (ER4).
- Manage factors that stress the crop or injure ears; ask the seed supplier about hybrids with ear-rot resistance, available for Aspergillus and Diplodia in some countries (ER3).
- If ear rot is significant, store kernels in a cool place; the fumonisin toxicosis outbreak cited from India shows the health stake (ER1, ER3).

Maharashtra note

No Maharashtra-specific ear-rot recommendation was found in the sources reviewed.

5. Remedies and treatment

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Foliar or soil fungicides	Ear rots	-	-	--	No maize claim in the CIB&RC list and no verified recommendation; not recommended

IIMR's bulletin lists stenocarpella, Fusarium and Gibberella, Aspergillus and Penicillium ear rots with screening methods (IIMR); the pages with details were beyond the part of the document that could be retrieved. No registered fungicide with an ear-rot claim on maize is in the CIB&RC list.

6. Sources and citations

[IIMR] Hooda, K.S., Bagaria, P.K., Khokhar, M., Kaur, H. & Rakshit, S. (2018). Mass Screening Techniques for Resistance to Maize Diseases. ICAR-Indian Institute of Maize Research, Ludhiana. https://iimr.icar.gov.in/wp-content/uploads/2020/03/Bulletin-Mass-Screening.pdf

[ER1] Unveiling toxigenic Fusarium species causing maize ear rot: insights into fumonisin (PAU Ludhiana). Frontiers in Plant Science, 2025. https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2025.1516644/pdf

[ER2] University of Nebraska-Lincoln: EC1901 Corn Disease Profile III - Ear rot diseases and grain molds (supplementary, non-Indian). https://extensionpubs.unl.edu/publication/ec1901/pdf/view/ec1901-2012.pdf

[ER3] NC State Extension: Corn ear rots (supplementary, non-Indian). https://content.ces.ncsu.edu/corn-ear-rots

[ER4] UF/IFAS Extension: Corn ear rot identification and management (supplementary, non-Indian). https://blogs.ifas.ufl.edu/alachuaco/?p=4115

[CIB] Central Insecticides Board & Registration Committee (CIB&RC), Govt. of India: 'Major Uses of Pesticides (Registered under the Insecticides Act, 1968) - Fungicides', up to 30.09.2025, 'based on certificates issued' (Directorate of Plant Protection, Quarantine & Storage). Copy used: https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Earlier edition (31.03.2024) on the official portal: https://ppqs.gov.in/sites/default/files/fungicides_31.03.2024.pdf

17. Common Smut and Head Smut

Maize | Ustilago maydis (common smut) and Sphacelotheca reiliana (head smut) | Fungal (smuts)

At a glance. Common smut forms large grey-white galls on any aerial part; head smut converts the tassel and ear into black powdery masses after a hidden systemic infection. Neither is on IIMR's priority list, and no Indian recommendation or registered maize fungicide was found, so this chapter rests on international sources.

1. Identification

Symptoms

- Common smut (boil or blister smut): tumour-like or mushroom-like galls on ears, tassels, stalks and leaves that later release sooty black spores (SM1, SM2).
- Head smut: infection is systemic but is obvious only on ears and tassels in mid to late season; affected ears are round or teardrop-shaped and filled with black powder, with no silk or kernels (SM1, SM3).
- Leafy, abnormal growths can occur on infected ears and tassels (SM1, SM3).

Distinguishing features

- Common smut galls are thick, fleshy and mushroom-like; head smut ears show a tangle of vascular strands among powdery spores (SM1, SM3).

2. Causes and progression

- Common smut: Ustilago maydis infects through wounds on actively growing tissue and develops locally (SM1, SM2).
- Head smut: Sphacelotheca reiliana survives as spores in soil for several years and infects seedlings through the roots (SM3).
- Common smut is favoured by wounding and plant stress (SM1).

3. Favourable conditions

- Wounding of plants and plant stress favour common smut (SM1).
- Spores persisting in soil, and seed-borne spores, carry head smut into new fields (SM3, SM4).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Hybrids differ in resistance to head smut; use resistant hybrids (SM1).
- There are no management recommendations for common smut in field corn in the Nebraska guide (SM1).
- A European paper recommends carboxin-treated seed to avoid seed-borne contamination of soil and field checks about a month after flowering (SM4).

Maharashtra note

No Maharashtra-specific information was found.

5. Remedies and treatment

Biological and non-registered options

- TNAU's smut advice (collect smutted heads in bags and dip in boiling water) is for sorghum head smut and is not a maize recommendation.

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Carboxin seed treatment	Head smut (Europe)	As per label	-	--	No maize claim for carboxin in the CIB&RC list; not recommended here

The CIB&RC list gives carboxin claims only for wheat, barley and cotton, so seed treatment against smut on maize would be off-label in India.

6. Sources and citations

[SM1] University of Nebraska-Lincoln NebGuide G2223: Smut diseases of corn (supplementary, non-Indian). https://extensionpubs.unl.edu/publication/g2223/2014/html/view

[SM2] APS Education Center: Common smut of corn (Ustilago maydis) (supplementary). https://www.apsnet.org/edcenter/pdlessons/Pages/CornSmut.aspx

[SM3] Plantix crop-disease library: Head smut of maize (supplementary, non-university source). https://plantix.net/plant-disease/en/100051/head-smut

[SM4] Gindrat, D. (1995). Head smut of maize at our door (Switzerland; AGRIS record; supplementary). https://agris.fao.org/search/zh/records/647359ef2c1d629bc97b7801

[CIB] Central Insecticides Board & Registration Committee (CIB&RC), Govt. of India: 'Major Uses of Pesticides (Registered under the Insecticides Act, 1968) - Fungicides', up to 30.09.2025, 'based on certificates issued' (Directorate of Plant Protection, Quarantine & Storage). Copy used: https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Earlier edition (31.03.2024) on the official portal: https://ppqs.gov.in/sites/default/files/fungicides_31.03.2024.pdf

18. Seed Rot, Seedling Blight and Root Rot

Maize | Complex of seed- and soil-borne fungi (Fusarium moniliforme named for root rot in CIB&RC) | Fungal

At a glance. Poor stands from rotting seed and blighted or wilting seedlings, mostly in cold, wet soils. This is the one area where CIB&RC lists several registered maize seed dressings, so it is the best-covered chapter for chemicals.

1. Identification

Symptoms

- Seed rot, seedling rot, seedling wilt and seedling blight are the maize claims on the CIB&RC list (CIB).
- Himachal Pradesh advisory names seed rot and seedling blight as problems in moist, cold fields (ISH).
- ICAR root-rot advice (state not stated) lists a root rot managed by drenching and soil Trichoderma (ISQ).

Distinguishing features

- Downy mildew seedlings show striping and white downy growth, not rotting (IIMR).
- Pythium stalk rot occurs near the base later in the season (IIMR).

2. Causes and progression

- The sources reviewed do not name the causal fungi apart from Fusarium moniliforme for maize root rot in the CIB&RC bio-fungicide list (CIBB).

3. Favourable conditions

- Moist, cold soil at sowing (ISH); water logging (ISQ).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Use recommended varieties and avoid sowing in moist and cold fields (ISH).
- Avoid water logging (ISQ).

Maharashtra note

No Maharashtra-specific recommendation was found.

5. Remedies and treatment

Biological and non-registered options

- Trichoderma harzianum 2.0% WP (NBRI-1055): seed treatment at 20 g/kg seed for maize root rot (Fusarium moniliforme) and Fusarium wilt (CIBB).
- ICAR root-rot advice: Trichoderma at 10 g/kg FYM as soil application (ISQ) - a state advisory.

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Thiram 40% FS	Seedling blight	9.6 g a.i. = 24 ml per 10 kg seed; slurry 100 ml	-	V1	CIB&RC; seed dresser
Thiram 75% WS	Seedling blight	18.8-22.5 g a.i. = 25-30 g formulation (seed quantity unit not legible in the list)	-	V1	CIB&RC; follow the label (Himachal advisory states 4 g/kg, ISH)
Carbendazim 25% + Mancozeb 50% WS	Seed rot, seedling blight	7.5 + 15 g a.i. = 30 g per 10 kg seed	0.1 L slurry as printed	V1	CIB&RC; seed dresser
Sedaxane 15.27% + Fludioxonil 7.64% + Metalaxyl-M 3.06% FS	Seedling rot, wilt and blight	0.3 ml/kg seed in 10 ml slurry as printed	-	V1	CIB&RC; seed treatment at sowing
Mancozeb 75% WP (Dithane M-45)	Root rot, soil drench	3 g/L	-	V3	ICAR state advisory (ISQ); CIB&RC maize claim for mancozeb is leaf blight and downy mildew

Seed dressings are the registered route. Doses differ between sources (thiram 75 WS: 25 to 30 g on the CIB&RC list, with the seed quantity unit not legible, vs 4 g/kg in Himachal's advisory); the product label governs.

6. Sources and citations

[IIMR] Hooda, K.S., Bagaria, P.K., Khokhar, M., Kaur, H. & Rakshit, S. (2018). Mass Screening Techniques for Resistance to Maize Diseases. ICAR-Indian Institute of Maize Research, Ludhiana. https://iimr.icar.gov.in/wp-content/uploads/2020/03/Bulletin-Mass-Screening.pdf

[ISH] ICAR-IIMR / AICRP Maize - Himachal Pradesh state recommendations (disease management table; page dated 22-11-2024). https://iimr.icar.gov.in/?p=1168

[ISQ] ICAR-IIMR / AICRP Maize - State recommendations: maydis leaf blight, downy mildew, banded leaf and sheath blight, root rot (state not stated on the page). https://aicrpmaize.icar.gov.in/?p=1197

[CIB] Central Insecticides Board & Registration Committee (CIB&RC), Govt. of India: 'Major Uses of Pesticides (Registered under the Insecticides Act, 1968) - Fungicides', up to 30.09.2025, 'based on certificates issued' (Directorate of Plant Protection, Quarantine & Storage). Copy used: https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Earlier edition (31.03.2024) on the official portal: https://ppqs.gov.in/sites/default/files/fungicides_31.03.2024.pdf

[CIBB] CIB&RC: 'Major Uses of Bio-Pesticides (Registered under the Insecticides Act, 1968)', updated up to 30.09.2025 (bio-fungicides). Copy used: https://www.jau.in/storage/document/dynamic_document/residue_bio_pesticide_30092025_20251226_780.pdf

Appendix A. Maize claims in the CIB&RC fungicide list

All maize entries found in the CIB&RC list up to 30.09.2025 (fungicides and the one bio-fungicide), as printed. Verify against the live register before use.

Product	Claim on maize	Dose as printed	Dilution	PHI
Kresoxim-methyl 44.3% SC	Turcicum leaf blight, rust	250 g a.i. = 500 ml/ha	500 L	25 d
Mancozeb 75% WP	Leaf blight, downy mildew	1.5-2 kg/ha	750 L	not stated
Zineb 75% WP	Leaf blight	1.5-2 kg/ha	750-1000 L	not stated
Pyraclostrobin 20% WG	Turcicum leaf blight	500 g/ha	500 L	40 d
Metalaxyl 35% WS	Downy mildews (seed treatment)	700 g/100 kg seed	slurry 0.75-1 L/100 kg	3.5-4 months
Metalaxyl-M 31.8% ES	Downy mildew (seed)	0.76 a.i. / 2.4 as printed	-	3.5-4 months
Thiram 40% FS	Seedling blight	24 ml/10 kg seed	100 ml slurry	seed dresser
Thiram 75% WS	Seedling blight	25-30 g formulation (seed quantity unit not legible)	-	7-10 as printed
Azoxystrobin 18.2% + Cyproconazole 7.3% SC	Downy mildew, TLB, rust	printed 1% (see note)	500 L	52 d
Azoxystrobin 18.2% + Difenoconazole 11.4% SC	Blight, downy mildew	1 ml/L	500 L	26 d
Carbendazim 12% + Mancozeb 63% WP	Downy mildew, leaf blight	1000 g/ha	500 L	37 d
Carbendazim 25% + Mancozeb 50% WS	Seed rot, seedling blight	30 g/10 kg seed	0.1 L slurry as printed	seed dresser
Fluxapyroxad 167 g/L + Pyraclostrobin 333 g/L SC	TLB, common rust	300 ml/ha	500 L	41 d
Mefentrifluconazole 200 g/L + Pyraclostrobin 200 g/L SC	Northern and southern leaf blight	400-500 ml/ha	500 L	48 d
Pyraclostrobin 133 g/L + Epoxiconazole 50 g/L SE	Leaf blight	750 ml/ha	500 L	48 d
Tebuconazole 50% + Trifloxystrobin 25% WG	Leaf blight	350 g/ha	500 L	15 d
Sedaxane 15.27% + Fludioxonil 7.64% + Metalaxyl-M 3.06% FS	Seedling rot, wilt, blight (seed)	0.3 ml/kg seed in 10 ml slurry as printed	-	seed dresser
Trichoderma harzianum 2.0% WP (bio)	Root rot (Fusarium moniliforme), Fusarium wilt	20 g/kg seed	-	-

Appendix B. Cross-check log and remaining gaps

What was checked and changed

- TNAU pages for leaf blight, turcicum leaf blight, charcoal rot and charcoal stalk rot were re-read and match the text (T2, T3, T4, T5).
- The full text of the CIB&RC fungicide list (up to 30.09.2025) was re-read. All 18 maize entries in Appendix A match the source. No other maize entries exist. In particular, carbendazim 50 WP, hexaconazole, propiconazole, single tebuconazole and carboxin have no maize claim, and no product has a claim for banded leaf and sheath blight, Curvularia, gray leaf spot, late wilt, smuts or ear rots.
- The ICAR / AICRP state pages were re-read. The page cited as ISP is the Karnataka recommendation (its state was previously described as unstated); ISH is Himachal Pradesh; ISA is Andhra Pradesh.
- Correction: the Andhra Pradesh mancozeb 2.5 g/L / Dithane Z-78 3 g/L advice names E. turcicum, so it now sits in the turcicum leaf blight table, not the maydis table.
- Gaps filled from the ICAR pages: Karnataka advice for turcicum blight, common rust, downy mildew and charcoal rot (hexaconazole, mancozeb, early sowing, potash, rotation with paddy or sesame); Himachal advice for brown stripe downy mildew, brown spot and seedling blight; the root-rot drench.
- Gaps filled by new chapters: Curvularia leaf spot, brown spot, zonate leaf spot, gray leaf spot, Pythium stalk rot, late wilt, ear rots and mycotoxins, common and head smut, and seed rot / seedling blight.
- Model-class check: the maize classes gray leaf spot, common rust and northern leaf blight are all covered. Gray leaf spot was strengthened with timing and fungicide-trial evidence (supplementary), and Appendix C maps each class and a healthy-leaf class to look-alikes and chapters.
- IIMR's bulletin lists the national diseases as turcicum leaf blight, maydis leaf blight, post-flowering stalk rots and banded leaf and sheath blight, and the regional diseases as polysora rust, brown stripe, sorghum and Rajasthan downy mildews, late wilt and cyst nematode. All except cyst nematode are now covered.

Remaining gaps

- Only the first part of the IIMR bulletin (to about page 57, bacterial stalk rot) could be retrieved; its late wilt, ear-rot and later stalk-rot pages were not, so the late wilt and ear-rot chapters rely on Indian research papers and international extension guides, flagged as supplementary.
- Gray leaf spot, smuts and ear rots have no Indian ICAR / TNAU / MPKV recommendation in the sources found; they carry no chemical recommendations.
- Maize cyst nematode (an IIMR regional problem), insect pests (fall armyworm, stem borers) and viral diseases are outside this document.
- Maharashtra-specific data are limited to MPKV Rahuri / AICRP Rahuri material. No PDKV Akola maize page was found. Zone-specific hybrid recommendations should be checked against the current AICRP release list.
- The CIB&RC compilation used is a hosted copy; check the live register before use.

Appendix C. Image-classification guide for the three model classes

The table below maps the three maize classes used by an image model (gray leaf spot, common rust, northern leaf blight) and a healthy-leaf class to the chapters above, with the visual features and the look-alikes that most often cause confusion. Confirm field diagnoses by scouting for sporulation and checking leaves of different ages, because several lesions look alike in a single photo.

Class	Chapter	What it looks like	Most likely confusion	How to separate	Management pointer
Northern leaf blight (turcicum)	1	Long, cigar- or spindle-shaped grey-green to tan necrotic lesions, 2.5-15 cm, starting on lower leaves; dark spore fuzz below in humid weather (T3, T2).	Maydis leaf blight; gray leaf spot	Maydis lesions are small, numerous and vein-limited (IIMR); gray leaf spot lesions are narrow, grey and rectangular (GL1, GL4); northern blight lesions are the broad tapered ones.	Chapter 1 table: many V1 options; MPKV 2024 schedule for Maharashtra
Common rust	4	Raised, powdery, cinnamon-brown pustules on both leaf surfaces; later brown-black (T6, IIMR).	Polysora (southern) rust	Polysora pustules are smaller, orange-red to light cinnamon and mostly on the upper surface (IIMR).	Chapter 4 table: kresoxim-methyl, fluxapyroxad + pyraclostrobin, azoxystrobin + cyproconazole (V1)
Gray leaf spot (Cercospora)	13	Long, narrow, sharply rectangular grey to tan lesions running parallel to the veins, often starting on lower leaves; may coalesce (GL1, GL4).	Northern leaf blight; maydis leaf blight	Lesions are rectangular and held between veins, grey rather than tan, and narrower than northern blight lesions (GL4, IIMR).	Chapter 13: resistant hybrids and rotation; no maize label claim in CIB&RC
Healthy leaf	-	No necrotic lesions, pustules or sporulation. The sources in this document do not describe healthy maize.	Early disease, natural ageing of the lowest leaves, non-disease stress	General caution, not from a cited source: a few yellowing lower leaves can be natural ageing. Striping without necrotic lesions may be downy mildew (Ch. 6) or banded brown spot (Ch. 11).	Nothing to treat; scout for the features above

Coverage check: all three classes are covered (chapters 1, 4 and 13). Gray leaf spot was the thinnest, because no Indian ICAR, TNAU or state-university source was found; it now carries supplementary trial evidence from South Africa, Brazil and the US, labelled as research only.